

Employers and Unemployment Insurance Take-Up[†]

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We quantify the employer's role in unemployment insurance (UI) take-up. Employer effects on claiming and appeals are substantial, and those effects are negatively correlated, consistent with appeals deterring claims. Low-wage workers are less likely to claim and more likely to have their claims appealed than median-wage workers. Employer effects help explain these income gradients, so equalizing employer effects on claiming would increase the progressivity of UI. Finally, the main source of targeting error in UI is that eligible workers do not claim. (JEL J22, J31, J63, J64, J65)

The revival of interest in the role of employers in the labor market has focused mainly on employers' influence on pay (e.g., Abowd, Kramarz, and Margolis 1999; Card, Heining, and Kline 2013; Song et al. 2019; Lachowska, Mas, and Woodbury 2020). In this paper, we study a specific nonpay dimension of employers' behavior: their role in facilitating or hindering the take-up of a social insurance program, unemployment insurance (UI).

The employer's role in UI take-up is twofold. First, employers have an incentive to care whether workers collect benefits because they effectively pay for some of their laid-off workers' benefits through an experience-rated payroll tax; hence, they may act in ways that discourage workers from claiming.¹ Second, and relatedly, UI agencies involve employers in determining whether workers who claim are eligible for UI by asking them for information on the reason for separation and giving them the right to appeal a worker's claim. Indeed, a small industry is devoted to helping employers manage UI payroll taxes (in part by appealing claims). Policymakers have long expressed concerns about employers being too aggressive in using the

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¹Baicker, Goldin, and Katz (1998) give historical perspective on how the United States ended up with a UI system that combines federalism, experience rating, and limited benefit duration. The paper does not mention program take-up as a consideration in the early debates.

right to appeal.² Despite this institutional setting, work on UI take-up has emphasized worker-side factors, linking incomplete take-up mainly to UI benefit generosity (Anderson and Meyer 1997b), limited information about the program (Vroman 2009), and the hassle of claiming UI (Ebenstein and Stange 2010). More generally, work on program take-up rarely emphasizes the employer's role.³

In this paper, we use administrative data from Washington state from 2005 to 2013 to study the employer's role in UI take-up. We find evidence of employer-specific effects on whether laid-off workers claim UI benefits and the rate at which those claims are appealed. Moreover, we find that employer effects on claims and appeals are negatively correlated and that the employer effects help explain steep income gradients in claims and appeals. Thus, equalizing employer effects on claiming would increase the progressivity of the UI system. Finally, we find that the failure of eligible workers to claim UI is a more significant source of targeting error in the UI system than the receipt of benefits by ineligible workers.

We begin by describing the institutional setting for UI, which highlights the window into the UI take-up process offered by the data (see Section I). UI eligibility depends on three criteria: monetary eligibility (Does the worker have an adequate work history?), separation eligibility (Did the worker separate through no fault of their own?), and continuing eligibility (Is the worker able, available, and searching for work?). Workers who claim UI are asked to state a reason for their separation, and the UI agency routinely asks the employer about the conditions of separation. If the employer responds with a reason that differs from the worker's (or that may disqualify the worker), then the agency may gather additional information and formally determine or adjudicate the claim. We refer to this process of employer involvement as an appeal. The administrative data include the information used to determine monetary eligibility, whether the worker claimed UI, the worker's and employer's reasons for separation, and whether the worker ultimately received benefits. Thus, we observe the key relevant worker action (claiming UI) and the key relevant employer action (appealing the worker's claim).

Although we observe monetary eligibility reliably, separation eligibility is essentially unobservable; it is determined through a quasi-judicial process that depends on potentially unreliable information provided by workers and employers. Like Anderson and Meyer (1997b), we construct a sample of "likely eligible" separators (Section II) and (similar to Anderson and Meyer 1997b) find that about 45 percent of these separators in fact claim UI. We also find that employers disagree with the

²The Association of Unemployment Tax Organizations lists more than 20 companies that offer such services, and its employer web page says, "Many employers have found Unemployment Insurance cost management companies to be a cost-effective 'best practice,' in terms of administering UI claims and managing UI tax liabilities. Professional service organizations ... can help you minimize your unemployment insurance taxes and exposure" (<https://www.autax.org/employers.html>, last accessed February 6, 2023). The concerns of some UI observers were expressed in the report of the US Advisory Council on Unemployment Compensation: "Some members of the Council are concerned [that] ... under a system of experience rating, some employers might make excessive use of the appeals system" (US Advisory Council on Unemployment Compensation 1996, p. 19). For a journalistic account of the UI cost management industry, see Jason DeParle. 2010. "Contesting Job Loss Claims Becomes a Boom Industry." *New York Times*, April 3. <https://www.nytimes.com/2010/04/04/us/04talx.html> (last accessed July 5, 2022).

³More generally, few papers have discussed the role of employers in social program take-up; exceptions are Bana et al. (2023) and Aizawa, Mommaerts, and Rennane (2022), who study parental leave and workers' compensation take-up, respectively. Helpful general surveys of program take-up include Currie (2006) and Ko and Moffitt (2024).

reason for separation given by the claimant in 3.4 percent of claims. The wage gradients in both claims and appeals are steep: Workers in the bottom decile of pre-layoff hourly wages are 18 percentage points less likely to claim UI than workers in the fifth decile, and those bottom-decile workers are about twice as likely to have their claims appealed as those in the top 3 deciles.

To quantify the role of employers in claiming, we estimate two-way fixed effects models of claims and appeals using a sample of separators who were eligible to claim UI at least twice. Four pieces of evidence suggest that the assumption of exogenous mobility underlying our empirical model is approximately satisfied in this sample (see Section III). First, the sample consists of workers who switch jobs following a spell of unemployment, implying less scope for directed search—and thus endogenous mobility—than in the conventional setting, where all separators are used to study earnings (i.e., Abowd, Kramarz, and Margolis 1999—henceforth, AKM). Second, following Finkelstein, Gentzkow, and Williams (2016), we show that worker moves are approximately balanced in observable employer characteristics, suggesting that mobility is not likely based on unobserved employer characteristics. Third, following Card, Heining, and Kline (2013), we show that changes in worker-level outcomes are approximately linear and symmetric in changes among employers with different average employer claim and appeal rates. The approximate linearity of these relationships supports the additively separable specification of the two-way fixed effects model for claiming. Fourth, following Chetty, Friedman, and Rockoff (2014), we show that changes in employer effects are approximately unbiased predictors of changes in individual-level behavior, which suggests the two-way fixed effects model purges the employer effects of worker composition differences and match effects. Having established the approximate validity of the two-way fixed effects specification, we report a variance decomposition to show the presence of large employer effects on both workers' decisions to claim and employers' decisions to appeal.⁴ We find that employer effects are quantitatively important. The standard deviation of employer effects on claiming is about 15 percentage points (relative to a mean of about 50 percent). The ratio of the variance of employer effects to worker effects is larger for claiming UI than for wages. Employer effects on appeals are similarly important. Finally, if all employers with below-median employer effects on worker claims had effects at the median, then the UI claim rate would increase by 6 percentage points.

In Section IV, we study four implications of employer effects. First, employers with high claim effects and low appeal effects have other characteristics typically associated with more desirable employers, consistent with models of imperfect competition in the labor market. In particular, higher-claim and lower-appeal employers are higher paying. Consistent with unions facilitating access to UI, sectors with higher employer claim effects and lower employer appeal effects tend to have higher union density.

Second, consistent with employers using appeals to deter workers from claiming, employer effects on claims and appeals are negatively correlated: Workers are

⁴To address limited mobility bias, we report Kline, Saggio, and Sølvesten (2020) bias-corrected variance components.

less likely to claim UI when they are laid off by employers that appeal claims more frequently. Although the deterrent mechanism is plausible, other mechanisms could generate the negative correlation. For example, some employers might have more separations where eligibility is difficult to determine. Such heterogeneity could lead to lower claim rates and higher appeal rates, as we discuss below.

Third, employer effects account for well over half of the wage gradients in both claims and appeals. Given that employer effects are purged of worker characteristics such as workers' wages, this finding supports the important role of employer-specific effects on claims and appeals.

Fourth, employer effects also shape the progressivity of the UI system. To show this result, we consider a counterfactual in which we set all employer effects on claims to the average. This change increases the share of benefits received by workers with below-median wage rates by over 10 percent and makes workers in the ninth decile of the wage distribution (as well as those in the tenth) net payers into the system. As a result, the Gini coefficient of benefits received decreases by about 10 percent.

One alternative explanation for our results is that the employer effects on claims reflect differences across employers in the share of workers who are separation eligible, rather than in the claim rate of workers. Because separation eligibility is essentially unobservable, we undertake three sensitivity analyses to show that the patterns we observe reflect differences in the behavior of separation-eligible workers across either employers or the wage distribution (see Section III E). We first restrict attention to samples of workers who separate in mass layoffs. These workers are especially likely to have lost their jobs through no fault of their own and should be separation eligible. The striking income gradients in claiming and appeals persist in the mass-layoff sample. Moreover, by using various thresholds for a mass layoff, we show that the variance of employer effects on claiming is if anything larger in the mass-layoff sample than in the main sample.

Second, we add a different group of workers who are very likely to be separation eligible: workers on temporary layoff. The variance of employer effects on claims is again essentially unchanged, which suggests that our approach to estimating employer effects controls for the different types of separations that workers make.

Third, we add back workers who are very unlikely to be separation eligible. We examine first the subset of separators who made job-to-job transitions, then the sample of all monetarily eligible separators. The variance of employer effects is again unchanged by these additions.

In a conceptually distinct exercise, we test the sensitivity of the specification to the inclusion of the replacement rate. The concern here is that the employer effects could be correlated with workers' UI replacement rates, which have been shown to influence workers' claiming behavior (e.g., Anderson and Meyer 1997b). The variance of employer effects on claiming is unchanged by this inclusion; hence, the overall collage of evidence suggests that the results on the large variance of employer effects (as well as the presence of the wage gradient) are not driven by sample selection or model misspecification.

In Section V we develop a simple statistical framework to measure claims, appeals, and receipt rates among separation-eligible and -ineligible workers. The

key assumption is that the marginal workers induced into separating by a mass lay-off are separation eligible. This variation allows us to estimate claim and appeal rates among the separation eligible. We find that the main source of targeting errors in UI is that eligible workers do not claim UI, rather than that ineligible workers do. Hence, UI financing reforms that decrease experience rating and thus employers' incentives to appeal could decrease targeting errors.

A substantial literature has examined the effect of experience rating on hiring and separations; see, for example, Ratner (2013); Johnston (2021); Guo (2023); and Guo and Johnston (2021) for a review. But studies of the possible role of employers in limiting UI claims are surprisingly few, and the studies that have examined the role of employers have not documented employer heterogeneity in UI take-up. Anderson and Meyer (2000) found that, after Washington's UI payroll tax became experience rated in 1985, UI claim rates fell and denial rates increased. Based on interviews with job losers, Gould-Werth (2016) found that some employers actively help workers in claiming UI, whereas others are indifferent, and still others actively impede claims. Auray, Fuller, and Lkhagvasuren (2019) and Auray and Fuller (2020) used state-by-time variation in unclaimed benefits to study the effect of denied claims on UI take-up and link their results to experience rating. Van Doornik, Schoenherr, and Skrastins (2023) show that in Brazil (which lacks experience rating), employers strategically time layoffs: When employees become UI eligible, employers lay them off and then recall them after benefits are exhausted.

I. Institutional Setting

Two features of the UI program are central to take-up and receipt of benefits: (i) the eligibility criteria for UI, which include the worker's employment history, conditions of separation, and availability and willingness to search for employment; and (ii) the experience rating of the payroll tax that finances UI, which gives employers an interest in whether workers they lay off receive benefits and creates an incentive for employers to report (perhaps strategically) that a laid-off worker separated for a reason that would disqualify her.⁵ This section describes these features in turn.

A. Eligibility for UI Benefits

UI benefits are not automatically paid to laid-off workers. The worker needs to file a claim, and an administrative process determines whether the worker is eligible for benefits. Figure 1 illustrates this process of claiming and eligibility determination and indicates the aspects of the process that we observe in the data by nodes shown with bold outline.

⁵The employer might also report that the worker separated for a reason that would not disqualify the worker but would make the benefits paid "nonchargeable"—see the discussion below.

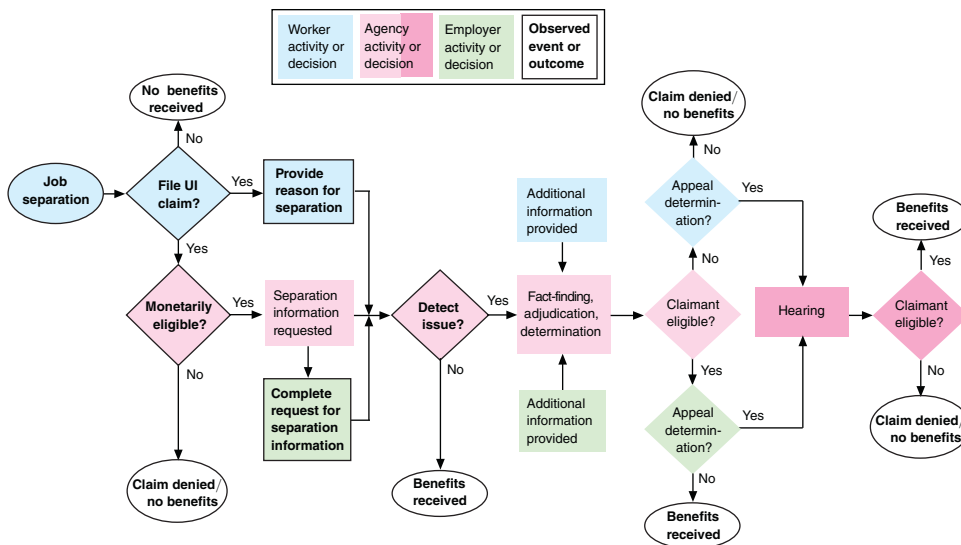


FIGURE 1. FLOWCHART SUMMARIZING THE CLAIMING, DETERMINATION, AND APPEAL PROCESS

Notes: This flowchart illustrates the process of claiming and eligibility determination for UI benefits described in Section IA. Events and decisions observed in the data are shown by nodes outlined in bold. Following job separation, a worker decides whether to claim benefits, and if she does, then the UI agency asks the worker questions about the conditions of separation and determines whether she is monetarily eligible (i.e., has an adequate work history to qualify). If the worker is monetarily eligible, then the agency requests information about the worker's conditions of separation from the employer(s). If the agency detects a separation issue (for example, due to the claimant's or employer's reasons for separation or if the claimant's and employer's reasons conflict), then the agency requests further information in order to adjudicate the issue and make a formal eligibility determination. The outcome of this determination can be appealed by either the worker or the employer, and the appeal is heard by an administrative law judge, who makes a final eligibility determination.

If a worker files a claim, then the state UI agency, in our case the Employment Security Department (ESD), determines the worker's eligibility for benefits based on three sets of criteria:⁶

- (i) *Monetary* eligibility: whether she has an adequate work history to qualify for benefits;
- (ii) *Separation* eligibility: whether she lost her job due to lack of work and through no fault of her own;
- (iii) *Continuing* eligibility: whether she is able, available, and searching for work.

⁶The claiming process is broadly similar among the states, but we refer to specifics of the process used in Washington, which are described in an extensive "Unemployed Worker Handbook" (Employment Security Department 2019). In 2013, most initial claims in Washington were filed either online (about 47 percent) or by telephone (51 percent), with most of the remaining claims filed by employers. The use of telephone filing is higher in Washington state than nationally: According to the US Department of Labor's Benefit Accuracy Measurement program, nationally in 2013, 63 percent of claims were filed online and 30 percent by telephone, with the remainder filed by employers, in person, or by mail. Ebenstein and Stange (2010) show that telephone and Internet claims largely replaced in-person claims between 1995 and 2005, but this apparently dramatic change had no appreciable effect on UI take-up overall or on the mix of claimants by previous earnings.

In Washington, the monetary eligibility requirement is at least 680 hours of work in the “base period,” which is approximately the year before the claim.⁷ The ESD determines monetary eligibility by referring to employers’ quarterly administrative wage and hour reports.⁸

To be separation eligible, the worker must have been laid off (that is, the job was eliminated and the worker will not be replaced), terminated for a reason other than misconduct (for example, the worker did not have the skills to perform the job), or quit for “good cause.”⁹ The worker does not need to have lost a job due to a mass layoff or employer downsizing. Indeed, in Section IIB, we show that about 40 percent of UI recipients separated from stable or growing employers.

To determine separation eligibility, the ESD relies mainly on information it obtains from the worker and the employer. When a worker files an initial claim, the ESD asks questions about why she became unemployed.¹⁰ If the responses raise an issue about the worker’s eligibility, then the agency may decide to perform a formal “determination” or “adjudication” of the claim, following up with a fact-finding letter and requesting further information (the letter is displayed in Supplemental Appendix Figure A2). The agency also routinely informs the worker’s base period employer(s) that the worker has claimed benefits and requests information about why the worker separated (see Supplemental Appendix Figure A3). The employer’s response may also trigger a fact-finding inquiry, for example, if the employer responds that the worker separated for a reason that would be disqualifying or if the employer’s and worker’s reasons differ in ways that raise a separation eligibility issue.

Either the claimant or the employer has the right to formally appeal the outcome of a determination to a separate state agency; in Washington, the agency is called the Office of Administrative Hearings. Appeals are conducted by an administrative law judge who hears testimony and evidence given under oath.¹¹ We do not observe whether a claim was formally adjudicated by the agency or appealed by either party. Rather, as we discuss in Section IIE, we observe when the reasons for separation given by the worker and the employer differed, which for our purposes is the relevant disagreement. Accordingly, when we refer to an appeal, we will be referring to

⁷This work must be for an employer covered by the UI system. The base period can be defined in either of two ways. The standard base period is the first 4 of the 5 completed quarters before the quarter in which a claim is filed. The alternative base period, for claimants who do not meet the 680-hour requirement in the standard base period, is the 4 completed quarters before the quarter of filing. Monetary eligibility is a misnomer in Washington because the state uses *hours* to determine eligibility. All other states use some measure of previous earnings, so we use this conventional terminology.

⁸Determining monetary eligibility requires additional inquiry and fact-finding by the agency if a worker’s hours have not been reported (or are misreported) in the wage records or if the claimant worked outside Washington during part of the base period.

⁹Washington currently has several good-cause reasons for quitting: sickness or disability, need to care for an immediate family member who is sick or disabled, a cut in usual pay or work hours by 25 percent or more, and moving with a spouse or partner who is relocating, among many. The criteria are fully described in Employment Security Department (2019) and Revised Code of Washington, Chapter 50.20 (“Benefits and Claims,” <https://app.leg.wa.gov/RCW/default.aspx?cite=50.20>, last accessed April 21, 2022).

¹⁰Screenshots of the questions Washington claimants must answer are shown in Supplemental Appendix Figure A1. The process of determining separation eligibility varies substantially among the states and is described and analyzed in Corson, Hershey, and Kerachsky (1986) and Fishman et al. (2003).

¹¹Typically, the judge reaches a decision within two to three weeks. If a worker receives benefits during an appeal, and the appeal goes against the worker, then the benefits must be repaid.

a case where the employer and the agency gave different reasons for separation, not to a case that went through the formal administrative appeal process.

The claimant must also be “able, available, and searching for work” (the continuing eligibility criteria). Workers answer questions about their availability and willingness to search when they initially claim, and to continue their eligibility, they need to keep a record of employer contacts and other job search activities.¹² For our purposes, the continuing eligibility criteria matter only at the time of the initial claim because we are concerned only with whether benefits are denied at that time.

The three eligibility conditions are not necessarily determined in a set order because issues may arise at any stage. Only after the agency has determined that the claimant meets all the eligibility criteria is a claim “certified,” after which the worker can expect to receive benefits within four to five weeks.¹³

B. Experience Rating and the Employer’s Role in Eligibility Determination

In the United States, UI benefits are financed by an experience-rated payroll tax, collected entirely from employers in most states. Under experience rating, each employer’s payroll tax rate increases as a function of the UI benefits paid to workers who separate from the employer. Although the schedule mapping an employer’s experience into a tax rate in Washington state is elaborate, it results in nearly full experience rating through much of its range, such that in expectation, taxes increase by \$1 for each \$1 of benefits charged to the employer.¹⁴ Washington’s schedule is capped and becomes flat at 5.4 percent, so at some point marginal experience rating is effectively eliminated; Supplemental Appendix Figure A4 illustrates the tax schedule.

Experience rating is a unique feature of the US system and had three main goals at its inception (Blaustein, Cohen, and Haber 1993). First, it acts as a layoff tax because in expectation some laid-off workers will collect UI; moreover, the size of the tax is countercyclical because workers who separate in recessions are likely to experience longer unemployment spells and collect more UI benefits (Duggan, Johnston, and Guo 2023). Second, it distributes the costs of UI “equitably” by charging more to employers who use the system more. Third, it provides incentives for employers to provide information to state UI agencies about the reasons for separation, improving the integrity of the system by reducing the likelihood that ineligible workers will receive benefits.

¹² A claimant may also be required to attend job search workshops and receive other employment services, including job referrals by the agency. (See <https://esd.wa.gov/unemployment/job-search-requirements> for a list of activities that satisfy the requirement. The activities include “prepare a 30-second ‘elevator speech’ to use at job fairs” and “set up an account at [worksourceWA.com](https://worksource.wa.com).”) On the effects of work search requirements in Washington State, see Lachowska, Meral, and Woodbury (2016).

¹³ Washington made the first UI payment within 28 days of the end of the week in which the initial claim was made for at least 90 percent of initial claims in most months during the time period we are examining. See <https://oui.doleta.gov/unemploy/btq.asp>.

¹⁴ Two details are worth mentioning. First, not all benefits are charged to the employer; for example, in some cases, benefits paid to workers who have quit voluntarily for good cause are not included in the experience rating calculation. Second, to calculate an employer’s experience, Washington uses a so-called benefit ratio formula, under which the tax rate depends on the ratio of benefits charged to taxable payroll in the previous four years. The taxable wage base in Washington state was \$30,500 in 2005 and \$39,800 in 2013. See Miller and Pavosevich (2019) and Lachowska, Vroman, and Woodbury (2020) for further discussion and analysis of experience rating methods. On the enforcement and administration of other tax systems, see the review by Slemrod and Yitzhaki (2002).

But in addition to these advantages, experience rating creates an incentive for employers to discourage workers from collecting UI. Indeed, in Supplemental Appendix A we show that variation among employers in claim rates has as much power to explain employers' payroll taxes as the more traditional explanatory factor of separation rates.

II. Data and Estimates of UI Claiming and Appeal Rates

In this section we introduce the data and describe how we estimate the UI claim and appeal rates. We then present summary statistics about how the claim and appeal rates vary by the base-period hourly wage.

A. Data Description

We have access to administrative wage and claim records for Washington from 2005 to 2013 (Washington State Employment Security Department 2002–2014). The wage records include the quarterly earnings and work hours for each worker-employer match, and the NAICS industry code of the employer.¹⁵ To ensure that we correctly identify the chargeable employer, we restrict the sample to monetarily eligible separations where the worker had only one base-period employer. The hours data tell us whether the worker satisfied the monetary eligibility criteria, and we define a *separation* as an event in which either (i) a worker's primary employer (the employer from whom work hours in that quarter were greatest) changes between two consecutive quarters or (ii) a worker has positive work hours in quarter t and no work hours in quarter $t + 1$.

The claim records contain the date the worker claimed UI, the weekly benefit amount, the benefits actually paid to the claimant, the reasons for separation given by the worker and by the employer, and basic demographic information. We link a claim to a separation when the claim occurred within one quarter of the separation.¹⁶

Figure 1 illustrates the claim process, with the steps we observe in the data indicated by the nodes outlined in bold. We observe job separation, whether the worker filed a claim and was monetarily eligible, the reason for separation the worker gave, the reason for separation the employer gave, and whether the claimant ultimately received benefits.

B. UI Receipt after Separation from Stable and Growing Employers

Our first goal is to estimate the UI claim rate: the percentage of UI-eligible separators who claim benefits. The claim records tell us the numerator of this rate (the number of separators who claim benefits), but we cannot directly observe the

¹⁵The employer is the entity from which UI payroll taxes are collected. Most employers have a single establishment and are also a firm. For multiestablishment firms, the employer usually comprises the establishments in the state that are in the same industry (so a firm operating establishments in different industries may appear as more than one employer). For firms with establishments outside Washington, we observe only the firm's activity in Washington.

¹⁶About two-thirds of the claims in the sample occurred in the same quarter as a separation, with the remaining third split between the quarter before a separation (in anticipation of a layoff) or in the quarter after (when a separation occurs late in a quarter or a worker delays filing).

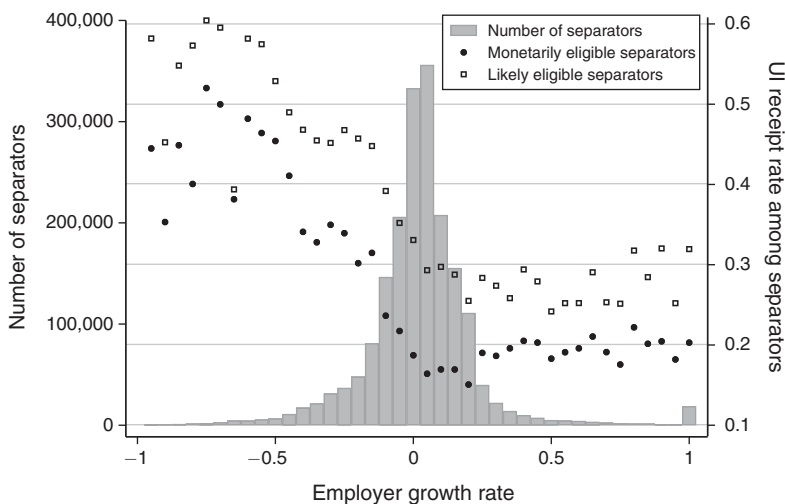


FIGURE 2. UI RECEIPT RATE OF SEPARATORS BY GROWTH RATE OF SEPARATING EMPLOYER

Notes: The solid black dots show the UI benefit receipt rate for all monetarily eligible separators (column 3 of Table 1), and the open squares show the receipt rate for likely eligible separators (column 4 of Table 1). On the horizontal axis, the employer growth rate is defined as the year-to-year percentage change in total employer hours. The histogram in the background shows the distribution of all monetarily eligible separators.

denominator (the number of UI-eligible separators) because it depends on both monetary and separation eligibility. Monetary eligibility can be observed accurately in the wage records, but the data contain no information on workers' conditions of separation unless they claim UI.

Because the separation eligibility requirement for UI is loss of a job due to lack of work and no fault of the worker, it might be presumed that all UI-eligible workers separate from employers that are contracting. If true, then this regularity would offer a convenient way to obtain the denominator of the claim rate because it would give a simple observable proxy for separation eligibility. This presumption turns out to be incorrect.

Using both employer-side and worker-side survey data matched to administrative data on employer growth rates, Davis, Faberman, and Haltiwanger (2012, Figure 6) and Flaaen, Shapiro, and Sorkin (2019, Figure 1B) show that layoffs are prevalent even at growing employers. We confirm this finding in our data by examining UI receipt rates for groups of separators, classified by the growth rate of employment at the employer from which they separated. The rationale for focusing on UI receipt (rather than eligibility) is that the administrative process has determined that these workers qualify for UI.

Figure 2 shows the results, with UI receipt rates for all monetarily eligible separators shown by solid dots and receipt rates for "likely eligible" separators shown by hollow squares. (These groups are described in the next section.) The figure shows that the UI receipt rate among likely eligible separators reaches a high of about 60 percent among workers separating from employers that close. The rate falls for workers who separate from employers that contract less severely, then reaches about 30 percent for workers separating from employers that are stable or growing.

Receipt rates for all monetarily eligible separators are lower than those for likely eligible separators, but the basic pattern is similar.

To summarize the data underlying Figure 2, about 40 percent of UI recipients in these two samples separated from employers that were either stable or expanding. The figure thus underscores the basic challenge in estimating the UI claim rate (that is, the proportion of UI-eligible workers who claim UI) and why we do not use employer contractions in our baseline definition of separation eligibility.

C. Estimating the UI Claim Rate

Because restricting attention to separations from contracting employers would exclude many UI-eligible job losers, we need to make alternative reasonable assumptions to identify workers who are likely to be eligible for UI.¹⁷ First, we drop separations that were likely voluntary because the worker quit to take another job. Specifically, we eliminate job-to-job transitions by dropping any separator who moved to a different employer in either the same or the following quarter and whose work hours decreased by at most 15 percent in the transition. (The hours decrease is relative to the quarter before the transition and corresponds to roughly a two-week reduction in work, during which it is unlikely a worker would claim UI.) Second, we eliminate separations that appear to have been withdrawals from the labor force by dropping separations that were followed by five or more quarters with zero work hours.¹⁸

These screens are imperfect because they do not necessarily eliminate all workers who separated voluntarily without good cause or were discharged for misconduct. For example, workers who quit to return to school, to search for full-time work, or to care for their children (if the children are not ill), and workers discharged for violating company rules, dishonesty, or repeated inexcusable absences would not be eligible for UI, but we cannot detect such cases. Thus, because the denominator includes ineligible workers, our estimates represent a lower bound on claim rates.

What share of our likely eligible sample is in fact eligible? Whether a worker meets the separation criteria for UI is inherently unobservable and cannot be known for workers who do not claim. Even for workers who claim, separation eligibility is determined with some error (see US Department of Labor 2020). Nevertheless, if we assume that the marginal workers who separate when an employer contracts are eligible for UI, then we can construct a measure of the share of likely eligible separators from stable employers who are in fact eligible. When we carry out this exercise, we find that about two-thirds of such workers are eligible (see Section V and Table 6).

How does the inability to directly observe separation eligibility affect our estimates of employer effects on claiming? If being discharged for misconduct or separating voluntarily are time-invariant characteristics of workers, then employer effects will be purged of these factors. Hence, employer effects will be estimated using variation

¹⁷ Our approach builds on Anderson and Meyer (1994, 1997b), as well as work by Bjelland et al. (2011); Hyatt et al. (2014); and Sorkin (2018). We do not know whether a claimant was able, available, and searching for work.

¹⁸ Using a five-quarter period of inactivity avoids dropping seasonal workers.

in claim rates among eligible workers. If they are not time-invariant characteristics of workers, then employer effects will also reflect differences in the probability of separation-ineligible claimants separating and evading the screens just described. We address this possibility in four ways. First, we show in Section IIF that our descriptive patterns persist in a sample of mass layoffs. Second, in Section IIID, we show that including workers who are likely ineligible has little impact on our estimates, suggesting that worker effects capture the relevant heterogeneity. Third, we show in Section IIID that our estimates of employer effects are, if anything, larger in mass-layoff samples. Finally, because marginal workers who separate from a contracting employer are more likely to satisfy the separation eligibility criteria than workers separating from a stable employer, in Section V we use the employer contraction rate as an instrument for identifying separation-eligible workers. This approach shows that the estimates of employer effects likely represent differences in claim rates among separation-eligible workers.

Because we define separations to exclude workers on temporary layoff who are recalled, we have excluded UI recipients on temporary layoff from both the numerator and denominator of the claim rate calculation. Temporary layoff unemployment was the focus of early classic work on the effects of incomplete experience rating on unemployment (notably Feldstein 1978 and Topel 1983). As we discuss in Section IIID, our main conclusions about employer effects change very little when we include them as UI claimants.

D. Incompleteness of UI Claiming

Table 1 shows summary statistics for various samples of workers, separators, and claimants. The table is organized so that the top panel summarizes means and standard deviations of worker-level characteristics and the bottom panel summarizes employer-level variables. In this section, we discuss the first four columns and return to the remaining columns after we introduce our analysis samples in Section III.

The first two columns of Table 1 show summary statistics for all worker-quarter observations (column 1) and for worker-quarter observations that would result in a monetarily eligible UI claim if the worker separated and claimed in that quarter (column 2). Conditioning on monetary eligibility reduces the sample size by more than one quarter. Workers who would be monetarily eligible for UI if they separated earn more, are employed by larger employers, and have accumulated more tenure with their employers.

Columns 3 and 4 of Table 1 show estimates of UI claiming. Among monetarily eligible separators, about 29 percent of separations result in a UI claim (column 3). Nearly two-thirds of separations are followed either by a quick transition to a new job or to exit from the labor force. These separations are likely not eligible for UI either because they are not associated with unemployment or because they would not satisfy the continuing eligibility criterion of being “able, available, and searching for work.” Dropping these separations increases the estimated claim rate to 45 percent (column 4). Relative to all monetarily eligible separators, workers who separate and do not make immediate transitions or exit the labor force have lower earnings and longer work hours.

TABLE 1—SUMMARY STATISTICS (MEANS WITH STANDARD DEVIATIONS IN PARENTHESES)

	Worker-quarter observations ^a		Monetarily eligible transitions ^b		Likely eligible separations of at least twice-eligible workers ^c		UI claims among at least twice-eligible workers ^d	
	All	Monetarily eligible in quarter	All	Likely eligible separations	Twice-eligible separators (switcher sample)	Leave-out connected set	Twice-eligible claimants (switcher sample)	Leave-out connected set
					(5)	(6)	(7)	(8)
<i>Worker-level variables</i>								
Claimed UI	0.020 (0.141)	0.023 (0.150)	0.288 (0.453)	0.453 (0.498)	0.474 (0.499)	0.505 (0.500)	1.000 (0.000)	1.000 (0.000)
Claim appealed (conditional on claiming)	0.021 (0.142)	0.021 (0.142)	0.031 (0.174)	0.034 (0.182)	0.030 (0.170)	0.033 (0.179)	0.029 (0.166)	0.031 (0.173)
Benefits received (unconditional)	0.015 (0.121)	0.018 (0.133)	0.220 (0.414)	0.350 (0.477)	0.363 (0.481)	0.396 (0.489)	0.791 (0.407)	0.809 (0.393)
Benefits received (conditional on claiming)	0.735 (0.441)	0.782 (0.413)	0.765 (0.424)	0.772 (0.419)	0.766 (0.424)	0.783 (0.412)	0.791 (0.407)	0.809 (0.393)
Benefits received (conditional on appeal)	0.583 (0.493)	0.624 (0.484)	0.611 (0.487)	0.618 (0.486)	0.592 (0.492)	0.636 (0.481)	0.611 (0.488)	0.655 (0.476)
Base-period earnings (\$2005) ^e	39,291 (78,026)	51,073 (50,332)	42,471 (46,024)	38,751 (39,911)	37,803 (34,861)	36,606 (36,479)	38,539 (29,303)	36,556 (36,430)
Base-period work hours	1,477 (793)	1,885 (457)	1,810 (516)	1,837 (545)	1,917 (559)	1,862 (556)	1,973 (516)	1,903 (526)
Base-period quarters	3.44 (1.01)	3.90 (0.369)	3.82 (0.495)	3.83 (0.498)	3.85 (0.463)	3.79 (0.546)	3.87 (0.435)	3.77 (0.559)

^aThe sample starts with 80.8 million worker-quarter observations in Washington from 2005:1 to 2013:4 in which the worker had the same employer (and only that employer) in the current and 5 previous quarters (column 1). (Employment in all 5 previous quarters is not required.) In 58.5 million of these quarterly observations, the worker had at least 680 work hours in either the 4 previous quarters (the standard base period) or the first 4 of the previous 5 quarters (the alternative base period) and hence would have been “monetarily” eligible for UI if s/he had separated in that quarter (column 2).

^bColumn 3 shows summary statistics for the 3.2 million job transitions observed among the 58.5 million worker-quarters in column 2. Of these 3.2 million transitions, we drop those where (i) the worker transitioned to another employer in either the same or the following quarter and experienced a decrease in work hours of no more than 15 percent in the quarter of transition or between the 2 quarters when the transition took place, and (ii) the separation was followed by 5 or more quarters with 0 reported work hours (an apparent labor force withdrawal). We refer to the remaining 1.0 million transitions (column 4) as “likely eligible separations,” and 45.3 percent of these resulted in a UI claim.

^cColumn 5 restricts the sample in column 4 to separations of workers who had two likely eligible separations during 2005:1 to 2013:4 but not within the same calendar year. These restrictions reduce the sample to about 142,000 worker-quarter observations. We refer to this as the “switcher” sample of separators. Column 6 restricts the sample in column 4 to the largest leave-one-out connected set. Column 6 is not a strict subset of column 5 because it includes workers who separated two or more times who are in the leave-out connected set and workers who separated twice in the same year (we do this to preserve connectivity), whereas column 5 includes workers who separated exactly twice in different years.

^dColumn 7 restricts the sample in column 5 to workers who separated and claimed twice during the period 2005:1 to 2013:4 but not within the same calendar year. These restrictions reduce the sample to about 21,000 worker-quarter observations. We refer to this as the “switcher” sample of claimants. Column 8 restricts the sample in column 5 to the largest leave-one-out connected set. Column 8 is not a strict subset of column 7 because it includes workers who claimed two or more times and possibly in the same year (again to preserve connectivity).

^e“Base period” refers to the standard base period (i.e., the first four of the five completed quarters before the initial UI claim).

(continues)

The separators in columns 3 and 4 are negatively selected relative to all monetarily eligible workers in column 2; e.g., they earn less, have shorter tenure, and work for lower-paying employers. Because we are interested in the experiences of workers who separate, for our purposes, it is fine that the sample of separators is meaningfully different from the sample of all workers.

TABLE 1—SUMMARY STATISTICS (MEANS WITH STANDARD DEVIATIONS IN PARENTHESES) (*continued*)

	Worker-quarter observations		Monetarily eligible transitions		Likely eligible separations of at least twice-eligible workers		UI claims among at least twice-eligible workers	
	All	Monetarily eligible in quarter	All	Likely eligible separations	Twice-eligible separators (switcher sample)	Leave-out connected set	Twice-eligible claimants (switcher sample)	Leave-out connected set
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Employer-level variables^f</i>								
Employer claim rate	0.399 (0.244)	0.404 (0.244)	0.434 (0.274)	0.458 (0.288)	0.448 (0.310)	0.478 (0.293)	0.616 (0.253)	0.597 (0.255)
Employer appeal rate	0.035 (0.102)	0.035 (0.100)	0.033 (0.105)	0.033 (0.109)	0.031 (0.106)	0.032 (0.102)	0.029 (0.102)	0.029 (0.089)
Employer separation rate	0.039 (0.077)	0.039 (0.080)	0.168 (0.239)	0.154 (0.222)	0.132 (0.198)	0.119 (0.183)	0.121 (0.173)	0.105 (0.154)
Employer earnings (\$2005)	39,291 (31,270)	46,466 (31,368)	43,155 (32,068)	41,055 (28,845)	42,490 (26,783)	43,890 (26,802)	45,991 (25,907)	47,029 (26,623)
Employer size (number of workers)	11 (862)	13 (971)	13 (211)	19 (264)	90 (482)	97 (502)	165 (739)	196 (819)
Number of workers	5,925,293	3,779,602	2,349,999	884,431	71,037	76,758	20,767	19,168
Number of employers	286,285	226,702	186,002	118,313	16,961	16,737	7,171	6,160
Number of worker-quarters	80,787,087	58,486,188	3,193,079	1,010,966	142,074	160,712	41,534	39,623

^f Employer-level variables are means taken for the worker's base-period employer and linked to the worker. The employer-level claim and appeal rates are computed as leave-one-out employer-year averages using the sample of separations in column 4. The employer-level separation rate is computed as the leave-one-out employer-year average using the sample of transitions in column 3. Employer-level quarterly earnings are computed as the average quarterly earnings of the base-period employer, using the sample in column 1. Employer size is computed as the average quarterly count of workers of the base-period employer, using the sample in column 1.

The claim rates in columns 3 and 4 are slightly higher than those found by Anderson and Meyer (1997b, Table 3) using similar samples, which makes sense because, although Anderson and Meyer (1997b) were examining a time period (1979–1983) when claim rates were generally higher, the six states they examined all tend to have lower UI reciprocity rates than Washington. The UI claim rate we estimate is similar to the estimate in the most recent Unemployment Insurance Nonfilers Supplement to the Current Population Survey.¹⁹

E. Appeals: Differences between Worker and Employer Reasons for Separation

As discussed in Section IA, when a worker claims, the UI agency requests information from the employer about why the worker separated. The employer's response may trigger a fact-finding inquiry if the employer's and worker's reasons differ in ways that raise a separation eligibility issue. Panel A of Table 2 shows the joint distribution of claims in Washington from 2005 to 2013, classified by the reasons for separation given by the claimant and the employer. We observe the claimant's

¹⁹ In the 2018 Nonfilers Supplement, the UI take-up rate was about 38 percent (i.e., UI applicants as a percentage of job losers and persons who completed temporary jobs who had worked in the 12 months before the survey); see US Department of Labor (2019, Table 3). The denominator likely includes some job losers who were not monetarily eligible, so the Nonfilers Supplement estimate likely understates the claim rate among those we consider likely eligible separators.

TABLE 2—CROSS-TABULATION OF REASONS FOR SEPARATION GIVEN BY THE EMPLOYER AND THE CLAIMANT

	Employer reason for separation					
Claimant reason for separation	Lack of work/ reduced hours	Voluntary quit	Discharge	Other	Not reported	Row sums
<i>Panel A. Joint distribution of claims, by reason for separation</i>						
Lack of work/reduced hours	752	5,939	9,080	293	239,994	256,058
Voluntary quit	71	84	43	11	44,237	44,446
Discharge	248	135	164	4	110,544	111,095
Other	81	138	69	23	7,929	8,240
Not reported/missing	0	0	0	0	37,859	37,859
Column sums	1,152	6,296	9,356	331	440,563	457,698
<i>Panel B. Proportion of claims on which benefits were paid, by reason for separation</i>						
Lack of work/reduced hours	0.93	0.51	0.69	0.67	0.94	0.75
Voluntary quit	0.73	0.46	0.42	0.45	0.33	0.48
Discharge	0.85	0.54	0.64	1.00	0.64	0.74
Other	0.90	0.41	0.55	0.61	0.71	0.64
Not reported/missing	n/a	n/a	n/a	n/a	0.73	0.73
Column means	0.85	0.48	0.58	0.68	0.67	0.65

Notes: Panel A shows the number of UI claims jointly classified by the reasons for job separation given by the employer and the claimant. The “Other” category includes leave of absence, partially employed, still employed, gross misconduct, labor dispute, and unknown. The sample is restricted to claims associated with “likely eligible” separations, defined as in column 4 of Table 1. When the claimant’s and employer’s reasons differ, we classify the claim as appealed by the employer (cells in bold), except when the employer’s reason is “lack of work/reduced hours.” In these cases, the employer may have placed the worker on standby (temporary layoff) or short-time compensation (work-sharing UI) and does not appear to be disputing the worker’s claim. The claims we refer to as appealed by the employer did not necessarily go through the formal appeal process described in Section IA; that is, our usage of the term “appeal” differs from the administrative usage. Calculations from Panel A: Probability claimant did not report a reason for separation = $37,859/457,698 = 0.083$. Probability employer did report a reason for separation = $(457,698 - 440,563)/457,698 = 0.037$. Probability that worker and employer gave different reasons (and the employer does not want the worker on standby) = $[(6,296 - 84) + (9,356 - 164) + (331 - 23)]/457,698 = 0.034$. When the employer reported a reason, the probability the employer reported “voluntary quit” or “discharge” and the claimant reported “lack of work/reduced hours” = $(5,939 + 9,080)/(457,698 - 440,563) = 0.88$. Panel B shows the share of UI claims resulting in payment of benefits, for each joint classification category of reason for job separation given by the claimant and the employer. For example, the claimant received benefits in 51 percent of the cases for which the worker reported lack of work/reduced hours and the employer reported voluntary quit as the reason for separation. Row and column means are claim weighted. See also the notes to Panel A.

reason for separation in more than 90 percent of the cases. In contrast, we observe the reason given by the employer in only about 4 percent of cases.

When the employer did give a reason for the separation, in about 93 percent of cases it was because the worker had given “lack of work/reduced hours” as the reason for separation, and the employer disagreed, usually saying the worker had quit voluntarily (35 percent of cases) or was discharged (53 percent of cases). *When a worker and an employer give different reasons for a separation (and the employer’s reason is not lack of work or reduced hours), we refer to this event as an appeal.*²⁰ This definition of an appeal differs from the definition of a formal appeal

²⁰ Table 2 shows that in a relatively small number of cases the employer reports “lack of work/reduced hours” as the reason for separation. The top of the form in Supplemental Appendix Figure A3 shows that the employer would report lack of work or reduced hours if it wants the worker on standby or is implementing short-time compensation (work sharing). In both cases, the employer expects the worker to receive UI, so we do not count these cases as appeals. Note that benefits paid to workers whose hours are reduced or who are placed on short-time compensation are chargeable, so an employer does not avoid benefit charges by reducing hours.

in Section IA in that it refers to an event earlier in the process of determining UI eligibility.

Panel B of Table 2 shows the percentage of claims for which benefits were paid under each of the circumstances shown. For example, when the worker said the separation was due to lack of work and the employer said the worker had quit voluntarily, the case was resolved in the worker's favor about 50 percent of the time. When the worker said the separation was due to lack of work and the employer said the worker had been discharged, the case was resolved in the worker's favor nearly 70 percent of the time.²¹

Because we observe the set of workers who claim, we know the denominator of the appeal rate. Table 1 shows that in 3.4 percent of the UI claims, the worker and employer gave different reasons for separation (and the employer did not give "lack of work/reduced hours" as the reason), which we have defined as an appeal.²² In more than 60 percent of these cases, benefits were paid to the claimant. Hence, from the employer's standpoint, most appeals are "unsuccessful" in that the employer's challenge to the claim was evidently rejected.

F. Wage Gradients in UI Claims and Appeals

Section IID showed that take-up is incomplete. In this section, we show that take-up is also unequal across the wage distribution. Figure 3 displays six binned scatterplots showing the relationships between base-period hourly wages and some key UI outcomes. Panel A plots the probability of claiming UI by workers' base-period hourly earnings. The small gray dots are quantiles based on the sample summarized in column 4 of Table 1. (The blue hollow circles denote deciles of the base-period hourly earnings.) The relationship has an inverted-U shape. At \$10 an hour, the claim rate is only 30 percent but rises steeply to about 50 percent between \$15 and \$20 an hour; workers earning the lowest hourly wage rates are least likely to claim. At wage rates greater than \$20 an hour, the claim rate gradually falls. This finding is similar to, for example, Shaefer (2010, Table 2, column 4), which shows that the claim rate among workers inferred to be eligible for UI follows an inverted-U shape, with the rate rising from the first to the fourth quintiles of the wage rate, then declining in the fifth quintile.

Panels B and C suggest that deterrence through appeals is one possible explanation for why the claim rate increases between \$10 an hour and \$15–\$20 an hour. The small gray dots in panel B show that low-wage claimants are substantially more likely than median- and high-wage claimants to have their claims appealed, so there is also a strong wage gradient in appeals. A negative wage gradient in appeals is also reported in Ashenfelter and Levine (2000), who study predictors of appeals

²¹ This table suggests ambiguity and confusion about what is meant by "discharge" under UI laws. Discharge can refer either to discharge for cause, in which case the worker is not eligible for UI, or to inability to perform the job, in which case the worker is eligible for UI. About a quarter of the time the worker said she was discharged, the employer did not appeal the claim, suggesting that these were cases where the worker was unable to perform the job and thus eligible for UI.

²² Anderson and Meyer (2000, Table 4) reports that 6.6 percent of claims were "separation issue denials" in Washington state from 1972 to 1997. Ashenfelter and Levine (2000, Table 1) reports that 1 percent of awarded claims in Wisconsin in 1994 were appealed by employers. These concepts of appeals differ from those in our data, so the numbers are not directly comparable.

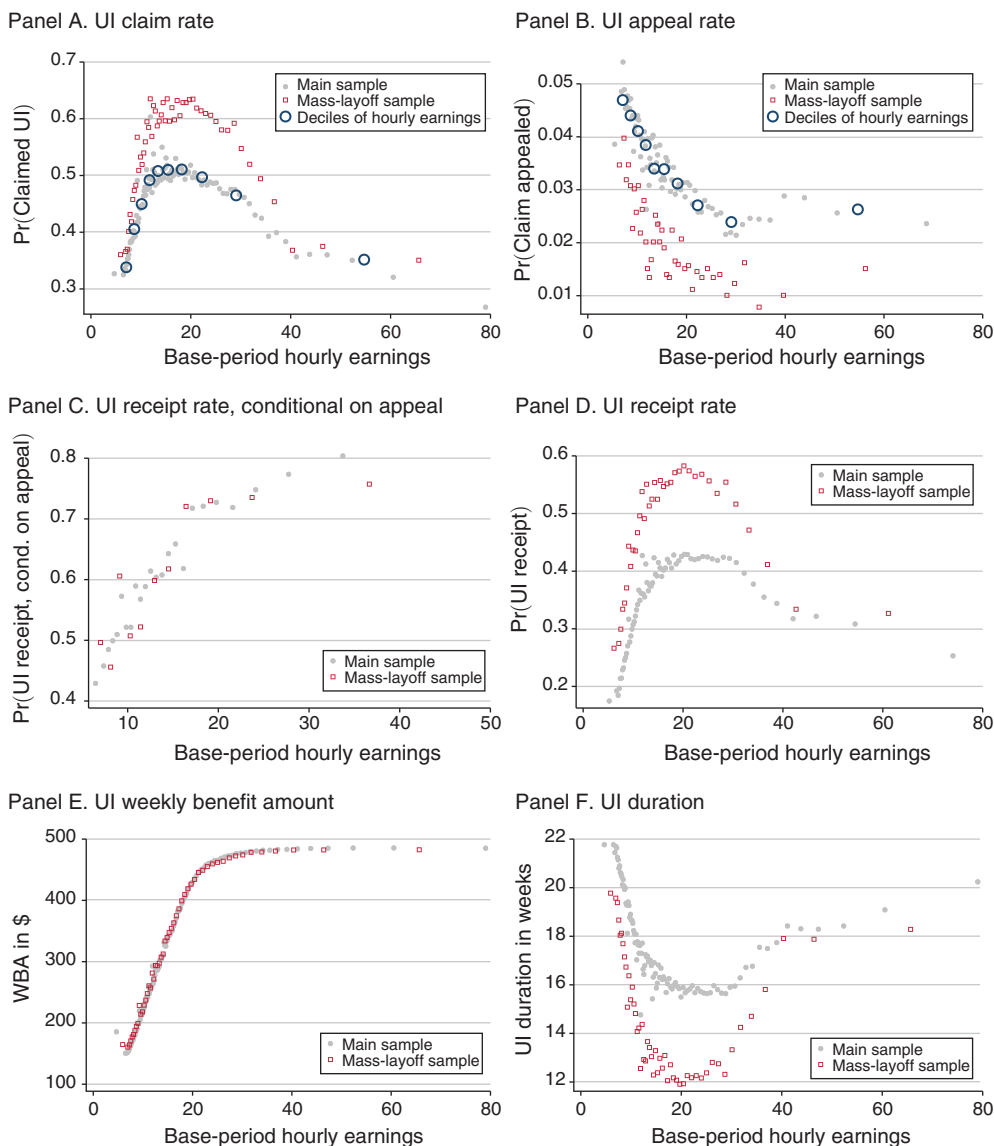


FIGURE 3. WORKER-LEVEL UI OUTCOMES, BY BASE-PERIOD HOURLY EARNINGS

Notes: The panels show scatterplots of the UI claim rate, appeal rate, receipt rate conditional on appeal, unconditional receipt rate, weekly benefit amount, and benefit duration, as a function of base-period hourly earnings. The small gray dots are based on the main analysis sample summarized in column 4 of Table 1. The hollow blue circles in panels A and B show claim and appeal rates by decile of base-period hourly earnings. The red squares are based on a subset of workers who separated from an employer whose total annual hours contracted by at least 20 percent and from which at least 5 workers separated in the current or adjacent quarters; this sample consists of nearly 138,500 separations. UI duration in weeks (panel F) is calculated as total benefits paid to the claimant divided by the claimant's weekly benefit amount, up to a maximum of 26 weeks. The number of points in the scatterplots is based on the cube root of the sample size.

of (continuing) claims using data from Wisconsin. Panel C shows that appeals of lower-wage workers are more likely to be decided against the claimant, which has not previously been reported. This finding has at least two interpretations. One is

that lower-wage workers who claim are more likely to be separation ineligible. Another is that lower-wage workers may be less likely to have the legal means or institutional understanding to present their case effectively.

Panel D combines the first three panels and shows that the likelihood of UI receipt follows a pattern similar to UI claiming displayed in panel A. The reason is that the likelihood of appeal is relatively low, so the likelihood of claiming dominates as an explanation for the wage gradient. Similar to panel A, this fact can be expected from the previous literature.

Panel E shows that, conditional on receipt of UI, the weekly benefit amount increases with the wage up to about \$20 an hour and is constant thereafter. This plateau at \$20 an hour occurs because workers reach the weekly maximum benefit amount at about this level of earnings. Accordingly, for workers with wage rates greater than \$20 an hour, both the replacement rate and the incentive to claim fall.

Finally, panel F shows the duration of UI spells among recipients. Conditional on receiving UI, lower-wage workers tend to have the longest UI durations, which suggests that low expected duration of UI receipt is an unlikely explanation for why low-wage workers have relatively low claim rates. This finding is similar to, for example, Moffitt (1985, Table 3), which shows that the UI exit rate (the inverse of duration) is increasing in the pre-separation wage.

One concern with these raw patterns is that they might reflect differences in separation eligibility by wage rate rather than differences in claiming, appeals, and other behavior by wage rate. To address this possibility, we generate the same plots using a “mass-layoff” subsample that is restricted to workers who separate in quarters when the employer contracts by 20 percent or more (denoted by red dots in the figure). This definition follows Jacobson, LaLonde, and Sullivan (1993), although we pick a less restrictive contraction rate to maintain sample size. The rationale is that, in mass layoffs, workers lose jobs because of employer-level distress, so workers are more likely to satisfy the separation-eligibility requirements for UI. The mass-layoff sample allows us to assess whether the patterns we observe in the main sample reflect differences in worker behavior by wage rates.

In general, Figure 3 shows that the wage gradients in the main sample also appear in the mass-layoff sample. Although some of the levels differ (e.g., claims are higher, and appeals are lower), the slopes of the wage gradients are similar. Accordingly, the gradients are unlikely to reflect differences in separation eligibility by wage rate. In Section V, we revisit the employer contraction rate as a source of variation and use it to more precisely quantify claim and appeal rates among the separation-eligible and -ineligible.

III. Employer Effects on Claims and Appeals

In this section, we estimate employer effects on claims and appeals. We first describe the empirical model. Second, we describe and implement several tests of the exogenous mobility assumption required to interpret the employer effects causally. Third, we quantify the importance of employer effects on UI claims and appeals. Finally, we show that previous employers have only a minimal effect on a worker’s claiming behavior, which suggests that the claiming environment of the current employer plays a key role in whether a separating worker claims benefits.

A. Empirical Model of Claims and Appeals

We model worker i 's decision to claim UI after separating from employer j in quarter t as follows:

$$(1) \quad c_{ijt} = \alpha_i + \psi_j(i,t) + \mathbf{x}'_{it}\boldsymbol{\beta} + \epsilon_{ijt},$$

where c_{ijt} equals one if the worker claims UI following separation from j and zero if the worker does not claim. We interpret the α_i as time-invariant worker-specific factors that affect whether i claims regardless of the employer. Worker-specific factors could include differences in knowledge, resources, propensity to quit voluntarily, propensity to be discharged for misconduct, or the returns to claiming implied by the wage level. The ψ_j represent the employer-specific claiming environment of employer j and reflect differences among employers that range from filing UI claims on behalf of workers to actively discouraging workers from claiming (see Gould-Werth 2016 for ethnographic evidence on these points). The $\mathbf{x}_{it}$ denotes year-quarter indicators that control for seasonal and business cycle fluctuations in the propensity to claim, and ϵ_{ijt} is an unobserved random error term.

Conceptually, it might make sense to include the replacement rate in the empirical model of claiming. For example, Anderson and Meyer (1997b) emphasize the role of the replacement rate in determining whether workers take up UI. Similarly, Supplemental Appendix Table A1 shows that in the cross section the replacement rate strongly predicts claiming behavior. Once we control for worker effects, however, the replacement rate has essentially no predictive power for claiming. Heuristically, in equation (1), the replacement rate is subsumed in the worker effects.²³ Supplemental Appendix Figure A5 provides intuition for why: The replacement rate schedule has a shallow slope such that even for large changes in wages, the replacement rate does not vary much within worker. Consistent with these results, we show in Section IIID that controlling for the replacement rate leaves our estimates of the variance of employer effects essentially unchanged.

To interpret ψ_j as an employer effect, we assume that workers do not sort to employers based on the error term, ϵ_{ijt} ; that is, mobility is exogenous. Informally, the assumption is that workers do not seek out employers where they are even more likely to be able to claim UI than would be implied by the employer effect. The analysis sample is restricted to workers who spent time in unemployment or nonemployment between spells and likely separated involuntarily. Because such workers typically have less scope to direct their search, we expect this type of sorting to be less likely than in a canonical AKM analysis of earnings, which typically includes all job transitions.

While exogenous mobility is often discussed as a restriction on worker behavior, because it is a restriction on equilibrium outcomes, it can equally be discussed in terms of employer behavior. In this context, the form of endogenous mobility that we might be worried about is how employers select workers to lay off (and to hire). As when framed in terms of worker behavior, the concern is that workers

²³Controlling for both worker and employer effects results in a coefficient close to zero; controlling only for employer effects generates a relationship fairly similar to the cross-sectional relationship.

are selected for layoff (or hire) based on the error term, ϵ_{ijt} . Sorting of workers to employers based on their fixed effects is not a violation of exogenous mobility. Indeed, it is consistent with exogenous mobility for different employers to select differently based on worker effects (e.g., some employers could select high- α_i workers, and others low- α_i workers).

If the exogenous mobility assumption fails, then the bias in the two-way fixed effects estimator of the variance components depends on the nature of the endogenous mobility. One natural intuition is that endogenous mobility would tend to compress the estimated variances and covariances of the worker and employer effects. To see this intuition, suppose there are match effects and workers move on the basis of the combined value of the employer effects and match effects. Endogenous mobility generates a negative correlation between match effects and employer effects because workers are willing to trade off between them by, for example, selecting a low-effect employer with a high match effect. This negative correlation implies that the distribution of employer effects plus realized match effects will be compressed relative to the unselected distribution.²⁴ Similar reasoning suggests that because endogenous mobility generates “mean reversion” in estimated employer effects, the estimated correlation between the employer effects on claims and appeals would be biased toward zero.

To estimate equation (1), we use the sample of likely eligible separators, shown in column 4 of Table 1. Worker effects are identified only for workers who meet the separation criteria in column 4 of Table 1 *more than once*. Also, we use the Kline, Saggio, and Sølvesten (2020)—henceforth, KSS—leave-one-out estimator, so employer effects are identified only for employers in the leave-one-out connected set of employers. This is the set of employers that remains connected by worker mobility after leaving out worker i separating from employer j in quarter t .²⁵

Column 6 in Table 1 characterizes the separators in the leave-one-out connected set. The restrictions implied by the leave-one-out estimator cut the sample size substantially, but on average, the workers do not differ greatly from the larger set of likely eligible separators in column 4. Specifically, the claim rate in column 6 is slightly higher than in column 4, and workers earn slightly more and work slightly longer hours.²⁶ These similarities suggest it is reasonable to extrapolate from the workers who were likely separation eligible more than once to the broader set of all separation-eligible workers.

The bottom panel of Table 1 shows that the leave-one-out set of employers is fairly similar to the employers of all likely eligible separators. Specifically, comparing columns 4 and 6, the average claim rate is slightly higher, the appeal rate is slightly lower, and average earnings are slightly lower for employers in the leave-one-out connected set than for employers of all likely eligible separators. Two quantitatively large differences do appear: The separation rate is meaningfully lower

²⁴To separate employer effects and match effects, we adopt the normalization that across all workers the distribution of match effects at each employer (“the unselected distribution”) is mean zero. Hence, the unselected distribution of employer plus match effects is simply the true distribution of employer effects.)

²⁵Any employer associated with a single mover is dropped from the leave-one-out connected set.

²⁶Column 5 in Table 1 shows the subset of separators who separated exactly twice but not in the same year; we refer to this subset as the switcher sample of separators. (The separations in column 6 may occur within the same year, in order to preserve connectivity.) The switcher sample is used in the exogenous mobility tests discussed in the next section. The characteristics of these switchers are similar to those in column 6.

(by about 20 percent), and employer size is larger by more than a factor of 5 among employers in the leave-one-out connected set. Nevertheless, the similarity of the claim rates suggests that it is reasonable to extrapolate from the leave-one-out sample of employers to the broader sample.

We structure the analysis of employer effects on appeals in a parallel way. Specifically, we restrict the sample to workers who claimed UI more than once and therefore could have experienced more than one appeal. Using the resulting leave-one-out set of employers, we estimate the analog of equation (1) using an indicator for whether a claim was appealed as the outcome variable. Column 8 of Table 1 shows summary statistics of this sample.²⁷ The notable differences between the samples in columns 6 and 8 are that the employers in column 8 tend to be much larger, the claim rate is higher, and the appeal rate is slightly lower.

B. Three Tests of the Exogenous Mobility Assumption

In this section, we describe three tests of the plausibility of the exogenous mobility assumption. The first test follows Finkelstein, Gentzkow, and Williams (2016) and looks at balance in the distribution of changes in employer characteristics among twice-separating workers (“switchers”). The intuition is that if workers select employers on the basis of the error term in claims, then they likely also select employers on the basis of the employer effects on claims or other employer characteristics. To the extent we do not see evidence of selection on employer characteristics, selection on the basis of the error term is less likely.

The second test follows Card, Heining, and Kline (2013) and looks at whether changes in outcomes when workers move between employers with high and low claim rates are symmetric. The intuition is that if workers move on the basis of the error term in claims, then they would only move to a lower-claim employer if they had a favorable draw of the error term. If so, then the change in the claim rate when moving between employers with low and high claim rates would be larger than the change when moving between employers with high and low claim rates.²⁸ In contrast, if mobility is exogenous, then we expect symmetric changes in the claim rate. The observable implication is that the relationship between the changes in the employer claim rate and the changes in the worker claim rate should be approximately linear. Linearity implies that the same change in employer-level claiming has the same effect on worker outcomes throughout the distribution. Accordingly, we test whether the linear specification fits the data better than a nonlinear (i.e., quadratic) specification.

Our implementation of the test in Card, Heining, and Kline (2013) differs from theirs in two ways. First, rather than providing a formal test, they plot the changes in earnings among workers who move between firms at the top and bottom quartiles of the firm earnings distribution, then visually inspect for symmetry. Second, they study earnings rather than claims, so the outcome is observed multiple times within

²⁷Column 7 in Table 1 shows the subset of separators who claimed exactly twice but not in the same year, that is, the switcher sample of claimants. This sample is used in the exogenous mobility tests discussed in the next section. The characteristics of this set of claimants are similar to the set of claimants in column 8.

²⁸This test has some limitations, which are discussed in Bonhomme, Lamadon, and Manresa (2019, p. 707).

a match, and they can examine pre- and post-trends. In contrast, we cannot examine pre- and post-trends for our outcome because so few workers separate and are likely eligible four or more times. (Moreover, we have vanishingly few cases where consecutive separations are from the same employer.)²⁹

The third test is similar to the second but replaces changes in employer claim rates with changes in employer claim effects. Doing so allows us to test whether changes in employer claim effects are an unbiased predictor of changes in worker outcomes. As emphasized by Chetty, Friedman, and Rockoff (2014), bias could stem from employer claim rates being contaminated by match effects or differences in worker composition among employers. Testing whether the coefficient on the change in employer claim effects (in predicting the change in worker outcomes) equals one tells us whether we have successfully purged the employer effects of match effects and worker composition differences. Nevertheless, noise in the estimated employer effects will attenuate the estimated relationship. To address this, we randomly split the sample of switchers to generate two estimates of employer effects and use one set of employer-effect changes to instrument the other.

C. Results from Tests of the Exogenous Mobility Assumption

In Section IIIA, we discussed how constructing a sample of workers who have a spell of unemployment between employers reduces concerns about violations of the exogenous mobility assumption. We now discuss the results of the three empirical tests.

Figure 4 shows the results of the first test. Among switchers, the figure plots the distribution of changes in employer claim and appeal rates (panels A and B) as well as three other employer characteristics: the employer separation rate, mean earnings, and size (panels C, D, and E). The histograms show that the distributions of moves are approximately symmetric around zero for all five of these changes in employer characteristics. Thus, workers do not appear to be moving on the basis of mean employer characteristics, and the exogenous mobility assumption appears to be valid.

Figure 5 and Table 3 show results of the second test, obtained by regressing changes in worker outcomes on changes in employer outcomes. The changes in worker outcomes are approximately linear in the changes in employer-level means, which supports symmetry.³⁰ Columns 2 and 5 of Table 3 show that a quadratic term is statistically insignificant, consistent with a linear relationship.³¹

Columns 3 and 6 of Table 3 show the results of the third test, where we regress the change in the probability of a claim (or appeal) on the change in estimated

²⁹For reference, we plot the figure that is closest to theirs in Supplemental Appendix Figure A6. Card, Heining, and Kline (2013) present two versions of their event study. The first is analogous to what we do in the next section and uses “raw” employer means without trying to adjust for worker composition (Card, Heining, and Kline 2013, Figure V). Specifically, we bin employers into quartiles by their average claim rates and examine the same subset of cross-quartile moves. In the second version of the event-study figure (Card, Heining, and Kline 2013, Figure VII), which we do not implement, they group employers into quartiles using the estimated employer effects.

³⁰The employer-level means are shrunken using a procedure described in Supplemental Appendix B.

³¹In Supplemental Appendix Figure A6, the event-style plots by quartile of claims, which parallel Card, Heining, and Kline (2013, Figures V and VII), also show symmetry in changes.

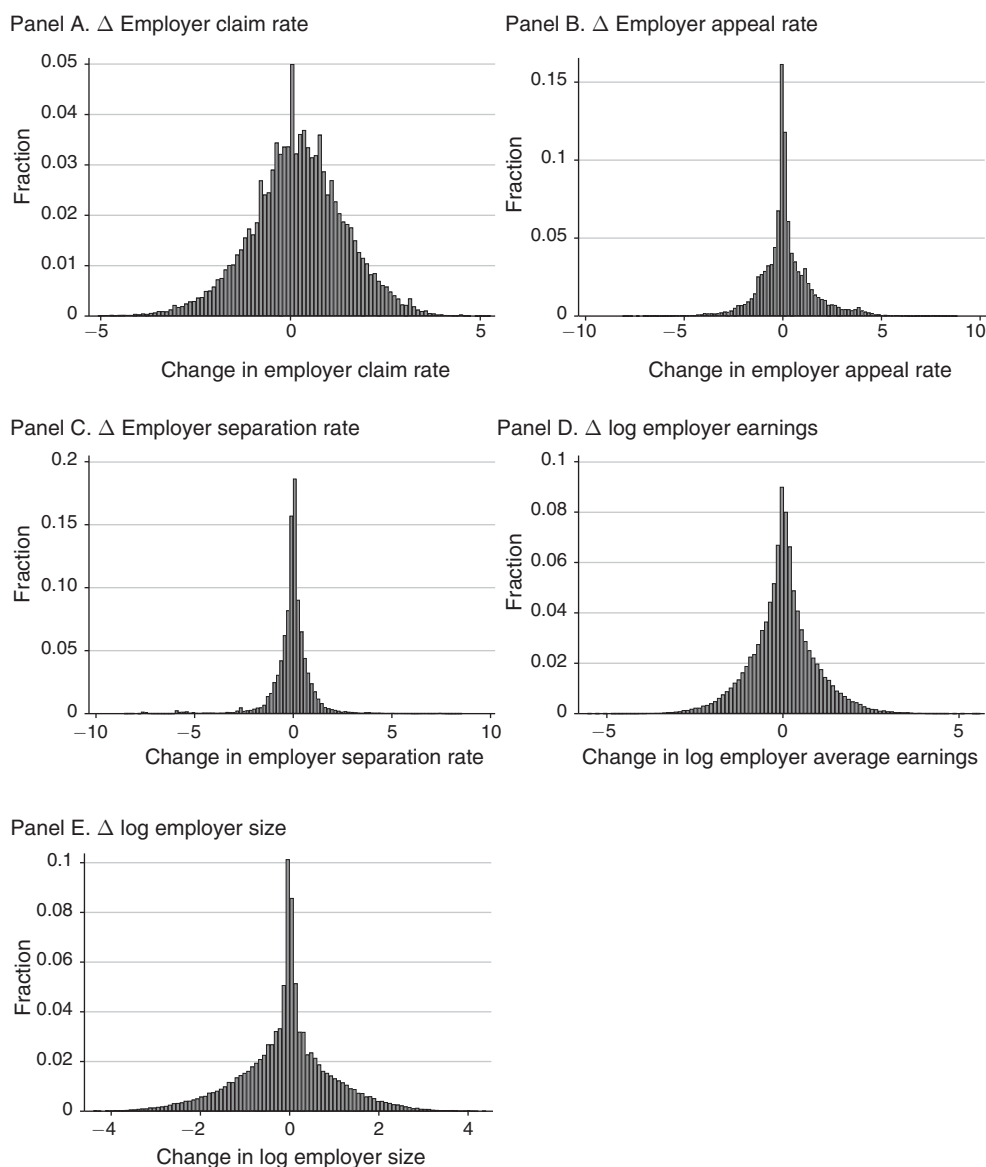


FIGURE 4. DISTRIBUTIONS OF CHANGES IN EMPLOYER-LEVEL CHARACTERISTICS AMONG SWITCHERS

Notes: The panels of this figure show the distributions of changes in employer-level UI claims (panel A), appeal rates (panel B), separation rates (panel C), average earnings (panel D), and size (panel E) for switchers (likely eligible separators who separated exactly twice but not in the same year). The distributions are based on the switchers in column 5 of Table 1 except in panel B, which is based on the sample of switchers summarized in column 7 of Table 1 (likely eligible separators who claimed exactly twice but not in the same year). The employer-level rates are computed as leave-one-out employer-year averages shrunk using the procedure described in Supplemental Appendix B.

employer effect on claims (or appeals). To address measurement error, we split the sample of movers in half to generate two estimates of employer effects, then instrument one with the other (see columns 1 and 2 of Supplemental Appendix Table A2 for the first-stage regression). In column 3 of Table 3, the 95 percent

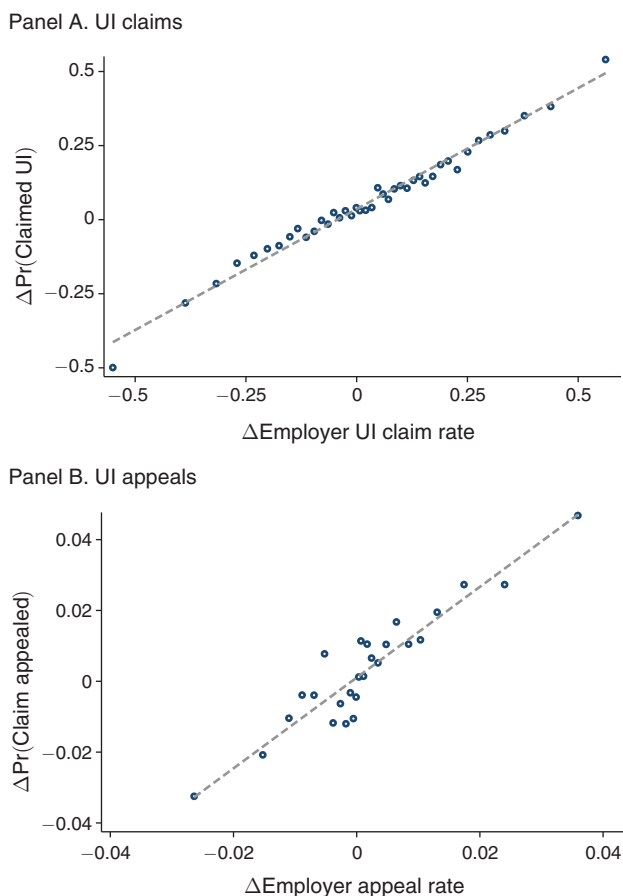


FIGURE 5. SWITCHER ANALYSES OF CHANGES IN UI CLAIMS AND APPEALS

Notes: For switchers (likely eligible separators who separated exactly twice but not in the same year), the figure shows changes in the probability of claiming UI (panel A) and having a claim appealed (panel B), against changes in the employer-level claim rate or appeal rate. The sample of switchers in panel A is described in column 5 of Table 1, and the switchers in panel B are defined in column 7 of Table 1. The light gray lines show univariate regressions of the change in the claim rate (or appeal rate) on the change in the employer claim rate (or appeal rate); see columns 1 and 4 of Table 3 for the underlying regression estimates. Both employer-level rates are computed as leave-one-out employer-year averages shrunk using the procedure described in online Supplemental Appendix B. The number of points in the scatterplots is based on the cube root of the sample size.

confidence interval of the estimated coefficient on the change in the employer effect contains one, implying that employer effects are approximately unbiased predictors of changes in worker outcomes and that the estimated employer effects are purged of match effects and worker composition differences.³²

In summary, the switcher analysis implies that both the worker's decision to claim UI and the appeal process satisfy the exogenous mobility assumption and are

³²The F -statistic for the first-stage relationship underlying the regression in column 6 is close to zero, making the point estimate of the coefficient on the change in the employer effect unreliable.

TABLE 3—SWITCHER ANALYSIS OF CHANGES IN UI CLAIMING AND APPEALS

Predictors	$\Delta\text{Pr}(\text{Claimed UI})$			$\Delta\text{Pr}(\text{Claim appealed})$		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\text{Employer UI claim rate}$	0.816 (0.022)	0.818 (0.022)				
$\Delta\text{Employer UI claim rate squared}$		−0.050 (0.060)				
$\Delta\text{Employer UI claim effect}$ ($\Delta\psi_{\text{claims}}$)			0.984 (0.033)			
$\Delta\text{Employer appeal rate}$				1.281 (0.137)	1.309 (0.157)	
$\Delta\text{Employer appeal rate squared}$					−2.667 (5.373)	
$\Delta\text{Employer appeal effect}$ ($\Delta\psi_{\text{appeals}}$)						−3.638 (143.936)
Constant	0.036 (0.004)	0.038 (0.003)	0.010 (0.003)	0.001 (0.002)	0.001 (0.002)	−0.013 (0.424)
Mean UI claim rate	0.475	0.475	0.491	1	1	1
Mean appeal rate	0.030	0.030	0.0365	0.0285	0.0285	0.0369
Number of workers	71,037	71,037	25,647	20,767	20,767	7,525
Number of employers	16,961	16,961	12,004	7,171	7,171	4,493
Sample	Switcher sample of separators	Switcher sample of separators	Switcher sample of separators (split)	Switcher sample of claimants	Switcher sample of claimants	Switcher sample of claimants (split)
First-stage F -statistic			165			0.001
Adj. R^2	0.0871	0.0871	0.573	0.00412	0.00408	−0.000133

Notes: Columns 1 and 2 show estimates from a regression of the change in a separator's probability of claiming UI on the difference in the UI claim rates of the separating employers, using the switcher sample of separators (column 5 of Table 1). Column 3 shows estimates from a regression of the change in a separator's probability of claiming UI on the difference in the AKM claim effects of the separating employers. In column 3, the change in AKM employer effects is instrumented by split-sample IV. Columns 4 and 5 show estimates from regressions of the change in a claimant's probability of experiencing an appeal on the difference in the appeal rates of the separating employers, using the switcher sample of claimants (column 7 of Table 1). Column 6 shows estimates from a regression of the change in a claimant's probability of experiencing an appeal on the difference in the AKM appeal effects of the separating employers. In column 6, the change in AKM employer effects is instrumented by split-sample IV. Average employer claim and appeal rates are computed as leave-one-out employer-by-year averages and adjusted for measurement error using the approach described in Supplemental Appendix B. Δ denotes a change from quarter $t + 1$ to quarter t . Standard errors, clustered at the employer level, are in parentheses.

well approximated by an additive model with worker and employer fixed effects. Accordingly, it makes sense to estimate an AKM variance decomposition.

D. Variance Decomposition of Claims and Appeals

Table 4 shows the variance decompositions implied by equation (1) for the UI claim rate (column 1) and appeal rate (column 2). Both are computed using the KSS estimator, which corrects for limited mobility bias.

The variance of the employer effect on claims, 0.022, corresponds to a standard deviation of about 0.148 ($= \sqrt{0.022}$) and is large in three senses. First, relative to the sample mean take-up rate of 0.505, the sixteenth to eighty-fourth percentile range (plus or minus 1 standard deviation) corresponds to a UI claim rate range of 0.35 to 0.65. Second, if employers with below-median claim rates had the median

TABLE 4—VARIANCE DECOMPOSITION OF UI CLAIM AND APPEAL RATES

Outcome:	Pr(Claimed UI)	Pr(Claim appealed)	Pr(Claimed UI)			
	Main sample	Main sample	Main sample + temporary layoffs	All monetarily eligible transitions	Main sample + job-to-job movers	Main sample, controlling for replacement rate
Sample	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Main samples</i>						
Total variance	0.250	0.030	0.218	0.220	0.217	0.250
Variance components						
Worker effects	0.049	0.0006	0.062	0.049	0.040	0.050
Employer effects	0.022	0.0003	0.021	0.020	0.022	0.023
Former employer effects						
cov[worker, employer]	0.001	0.0006	0.000	−0.001	−0.003	0.000
cov[worker, former employer]						
SD of employer effects	0.149	0.016	0.144	0.141	0.149	0.150
Number employers	16,737	6,160	16,737	16,737	16,737	16,737
Number former employers						
Number worker-qtrs	160,712	39,623	275,675	918,684	533,422	160,712
Number movers	62,118	16,641	71,876	340,256	220,281	62,118
<i>Panel B. Alternative samples</i>						
Outcome:	Pr(Claimed UI)					
	Mass layoffs (≥ 5% contraction)	Mass layoffs (≥ 10% contraction)	Mass layoffs (≥ 15% contraction)	Mass layoffs (≥ 20% contraction)	D-AKM on D-AKM sample	AKM on D-AKM sample
Sample	(7)	(8)	(9)	(10)	(11)	(12)
<i>Panel B. Alternative samples</i>						
Total variance	0.246	0.244	0.240	0.230	0.250	0.250
Variance components						
Worker effects	0.053	0.081	0.189	0.189	0.053	0.049
Employer effects	0.026	0.035	0.199	0.264	0.019	0.021
Former employer effects					0.004	
cov[worker, employer]	0.005	−0.010	−0.135	−0.161	0.002	0.002
cov[worker, former employer]					−0.004	
SD of employer effects	0.160	0.187	0.446	0.514	0.139	0.146
Number employers	2,809	1,497	416	102	8,298	8,298
Number former employers					5,469	
Number worker-qtrs	17,239	8,759	2,469	548	113,988	117,313
Number movers	6,620	3,231	787	183	54,381	42,280

Notes: Each column shows a KSS-corrected variance decomposition for claim or appeal rates. Columns 1 and 2 are estimated using the main sample (defined in columns 6 and 8 of Table 1). Column 3 shows the decomposition of claims using the main sample augmented by temporary layoffs (i.e., any claim not associated with a separation). Column 4 shows the decomposition of claims using the sample of all monetarily eligible transitions (column 3 of Table 1). Column 5 shows the decomposition using the main sample augmented by any within-quarter employer-to-employer transition. Column 6 shows the variance decomposition of claims, again using the main sample, but with controls for the workers' replacement rate. (The replacement rate is calculated using the standard weekly benefit amount divided by base-period hourly wage rate scaled by 35 hours per week.) Columns 7–10 show variance decompositions for claims estimated from samples of workers who separated in mass layoffs. (The samples are subsets of the sample in column 6 of Table 1.) In column 7, the sample of separators comes from employers whose hours contracted by ≥ 5 percent between quarters $t - 1$ and t , and from which as least 5 workers separated in quarter $t - 1$, t , or $t + 1$. In columns 8, 9, and 10, the contraction of hours required for a mass layoff is ≥ 10 percent, ≥ 15 percent, and ≥ 20 percent, respectively. (The requirement of at least five separations is maintained in each case.) Column 11 shows the variance decomposition from the “dual-employer effects AKM” model (D-AKM) (D’Addario et al. 2023). Column 12 shows the variance decomposition from a standard AKM model of claims, with the sample restricted to the subset of employers used to estimate the D-AKM model in column 11.

claim rate, then claiming would increase by about 6 percentage points, or 12 percent.³³ Third, the employer component is large in relative terms: It is almost half the size of the variance of the worker component. Thus, relative to worker effects, employer effects play a larger role in explaining UI claiming than in explaining earnings and wages.³⁴

The variance components emphasize the systematic dimension of the claiming decision, but there is a large idiosyncratic dimension as well. In particular, the worker effects explain only 20 percent of the variance, which suggests that the within-worker persistence of claiming is weak. Indeed, the share of twice-eligible separators who claim twice (columns 5 and 7 of Table 1) is 29 percent (20767/71037). If claiming decisions were independent within worker over time, then we would expect the share of twice-eligible workers who claim twice to be 22 percent (0.474^2 ; based on the mean claim rate in column 5 of Table 1). Put differently, we find about 30 percent more persistence in claiming than we would expect from independence, which is consistent with the limited role of individual effects in explaining claiming.³⁵

Column 2 of Table 4 shows the decomposition for the probability of an appeal. The standard deviation of employer effects is $0.017 (= \sqrt{0.00029})$, which is large relative to the mean of 0.03. The employer component is also large relative to the worker component: The ratio of the employer to worker variance is 0.40—comparable to the analogous ratio for claiming, which is 0.45. Thus, the role of employer heterogeneity in explaining appeals is similar to its role in explaining claims.

E. Robustness of the Estimated Variance Decompositions

The basic measurement issue we face is that true separation eligibility is unobserved. We test our empirical approach by performing analyses in which we add in separators who should be separation eligible or separation ineligible. If we found that the variance of employer effects on claiming was sensitive to including these groups, then we might worry that the variance of the employer effects on claiming reflects variation in the share of workers who are separation eligible, rather than the claiming rate among the separation eligible.

Columns 3, 4, and 5 of Table 4 add different groups of presumptively separation-eligible or -ineligible workers to the main estimation sample (column 1). In column 3, we add temporary layoffs—workers whose employers have placed them on standby.³⁶ In column 4, we include the monetarily eligible separators who we dropped from the main sample because they appear to have made a job-to-job transition or left the labor force and hence were either unlikely to claim or likely ineligible. In column 5, we add the subset of these monetarily eligible separators

³³This calculation uses the normal distribution.

³⁴For example, Kline, Saggio, and Sølvssten (2020, Table 2) find that the variance of the employer wage effect is about one-fifth ($0.0240/0.1119 = 0.21$) the size of the variance of the worker wage effect. Similarly, Sorkin (2018, Table 1) finds that the variance of employer earnings effects relative to worker effects is $0.14/0.51 = 0.27$. Finally, using Washington state data Lachowska, Mas, and Woodbury (2020, Supplemental Appendix Table B2) find the variance of employer effects relative to worker effects is $0.123/0.309 = 0.40$ for log earnings and $0.053/0.247 = 0.214$ for log hourly wages.

³⁵Michaelides (2014) finds that job losers who previously claimed UI were between 9.6 and 26.5 percentage points more likely to claim UI than job losers with no previous claim.

³⁶Temporary layoffs all claim and are likely eligible, but they do not separate from an employer.

who made job-to-job transitions. In all cases, we fix the set of employers to be the same as in the main sample (column 1).

In all cases, the variance of the employer component is largely unchanged.³⁷ Because employers differ in the composition of separations and the use of temporary layoffs,³⁸ it is reassuring that our measurement approach is able to effectively filter out these differences and arrive at similar answers to the importance of employer effects on claiming. The robustness across these samples suggests that, although sample choices affect the claim rate, they do not affect conclusions about the employer's role in explaining claiming.

A second question is whether the estimates are sensitive to inclusion of the replacement rate in the claim equation. As discussed above (see Section IIIA and Supplemental Appendix Table A1), the strong relationship between claiming and the replacement rate is a rationale for including the replacement rate in the claim equation; if employer effects and the replacement rate are correlated, then omitting the replacement rate will bias the estimated employer effects. When we include worker effects in the claim equation, the relationship between claiming and the replacement rate weakens (see Supplemental Appendix Table A1). Thus, it is not surprising that column 6 of Table 4 shows that including the replacement rate leaves the variance of the employer effect unchanged.

Third, we have emphasized that variation among employers in claim rates could reflect variation in the separation-eligible share of workers, rather than employer effects on the claim rate. To check this possibility, we consider various subsamples of "mass-layoff" separators. As discussed in Section IIF, the idea is that workers who lose their jobs in a mass layoff are more likely to have separated due to lack of work and hence to be separation eligible for UI.

In columns 7 through 10 of Table 4, we apply different thresholds for whether a separation occurred in a mass layoff. In column 7, a separation is associated with a mass layoff if (i) we observe at least 5 separations from an employer in quarters adjacent to the worker's separation quarter and (ii) the employer experiences at least a 5 percent contraction in total hours worked in quarter t relative to quarter $t - 1$. In column 8, the employer is additionally required to experience at least a 10 percent contraction; in column 9, at least a 15 percent contraction; and in column 10, at least a 20 percent contraction. As we require larger contractions in total hours, the number of employers for whom we can identify employer effects falls dramatically.

The basic pattern is that the variance of the employer effects tends to be larger in these "mass-layoff" samples than in the main sample; in fact, the variance is larger with larger contractions.³⁹ These results are consistent with the idea that the variance of employer effects represents variation in claiming rather than variation

³⁷ One difference is that the variance of the worker effects increases when we add the temporary layoffs, which is mechanical because we are adding a subsample of workers with take-up rates of 100 percent.

³⁸ Supplemental Appendix Table A3 documents the presence of employer effects of being in each of these categories, which are at least as large as the employer effects on claims.

³⁹ Supplemental Appendix Table A4 shows the variance decomposition when we restrict the analysis to the sets of "mass-layoff" employers in columns 7–10 but instead use any separation occurring from those employers. Most of the increase in the variance of the employer effects comes from restricting to mass-layoff separators, rather than changing the sample of employers, because the increase in employer effects is much less steep in this table than the table in the body of the paper.

in separation eligibility. Indeed, if anything, these results suggest that the main estimates represent a lower bound on the importance of employer effects on claiming.

F. *The Role of the Previous Employer*

The two-way fixed effects model in (1) assumes that the decision to claim depends on the current employer and the worker. One broad class of threats to the specification is any form of state dependence, where the claiming environment of the worker's previous employer continues to influence the worker's claiming behavior at the current employer. Information is a specific form of this state dependence: If the previous employer's claiming environment imparted information to the worker about how to claim UI, then this information could affect claiming when separating from the current employer.

To quantify this possibility, we follow Di Addario et al. (2023) and estimate a version of the model in equation (1), which includes both the current employer and the previous employer. Specifically,

$$(2) \quad c_{ijt} = \alpha_i + \psi_{j(i,t)} + \lambda_{h(i,<t)} + u_{ijt},$$

where ψ s are the current-employer effects on claiming, $j(i,t)$ indexes worker i 's current employer at time t , the λ s are the previous-employer effects on claiming, and $h(i, < t)$ indexes i 's previous employer (or, if the worker is observed for the first time in t , their nonemployment status). We refer to the model in equation (2) as the “dual-employer effects AKM” (D-AKM) model.⁴⁰

The D-AKM model nests the standard AKM model in (1) by allowing the estimates of λ s to differ from zero. The D-AKM model, however, requires a lagged employment state called “nonemployment,” which may lead to employer connections that do not exist in the standard AKM model (see Di Addario et al. 2023, Supplemental Appendix C for a formal discussion of identification). In order to make meaningful comparisons between the variance decompositions from the standard AKM model and the D-AKM model, we restrict the sets of current employers to be the same in the two models.⁴¹

Columns 11 and 12 of Table 4 display variance decompositions from the D-AKM model (column 11) and the AKM (column 12) model, using a common sample of current employers. Comparing column 1 (the AKM decomposition from the main sample) with column 12, the variances of the employer effects are virtually identical in the two samples despite a large difference in the sets of employers used to estimate the models. The variances of the standard AKM employer effects and the D-AKM current-employer effects are similar, and the variance of the previous employer effect is very small. We conclude that the role of the previous

⁴⁰The model controls for year-quarter dummies and the number of quarters between separating from the current employer and the previous employer. The results obtained without controlling for quarters since separation are nearly identical.

⁴¹We find this set by iterating between the identified sets for the standard and dual AKM models until convergence.

employer—which potentially includes information on how to claim UI—is quantitatively small.

IV. Implications of Employer Effects

In the previous section, we showed that there are employer effects on claiming and appeals, and these employer effects reflect the role of the current employer. In this section, we study various implications of the presence of employer effects and draw out the relationship between the employer effects and other dimensions of the UI system.

A. Correlates of Employer Effects on Claims and Appeals

We begin by projecting employer effects on claims and appeals onto various employer-level characteristics one at a time. Figure 6 shows slope coefficients from regressions of employer effects on four employer characteristics. To make the units comparable across characteristics, we standardize each to have unit variance. Panel A shows the coefficients associated with the employer effects on claims, and panel B shows coefficients associated with the employer effects on appeals.

Panel A shows that higher-claim employers tend to have lower separation rates and higher earnings, suggesting they are more desirable employers.⁴² These findings are consistent with the implication of models of imperfect competition (e.g., Lang and Majumdar (2004) and Sorkin (2018, Section VI)) where, if amenities are a normal good and there is utility dispersion across employers, then some employers will offer higher utility to workers through both better amenities and higher wages. In this case, the amenity is the employer claiming environment. This positive correlation between social benefit take-up rates and employer pay is similar to that found in Bana et al. (2023).

Larger employers tend to have lower claim rates, and multiple mechanisms drive this correlation. Work on employer size-wage effects (e.g., Brown and Medoff 1989) suggests that larger employers would be higher paying, and so the logic in the last paragraph implies a positive correlation between size and claim rates. But to the extent that setting up an appeals infrastructure involves fixed costs (for example, setting up a human resources department or hiring an unemployment compensation claims management company), we might expect a negative correlation. The results are inconclusive about which mechanism dominates because the relationship, although negative, is imprecisely estimated.

The correlation of the claim rate with the sector-level unionization rate is positive but is also imprecisely estimated.⁴³ This finding is consistent with work by Budd and McCall (1997, 2004), which suggests that unions help blue-collar workers with claiming UI and addressing employer appeals. Further, by helping with appeals, unions could deter employers from appealing.

⁴²In order to include voluntary separations, we use all separators to compute this rate.

⁴³The sector unionization rate is calculated using Current Population Survey Outgoing Rotation Group data (National Bureau of Economic Research 2005–2013).

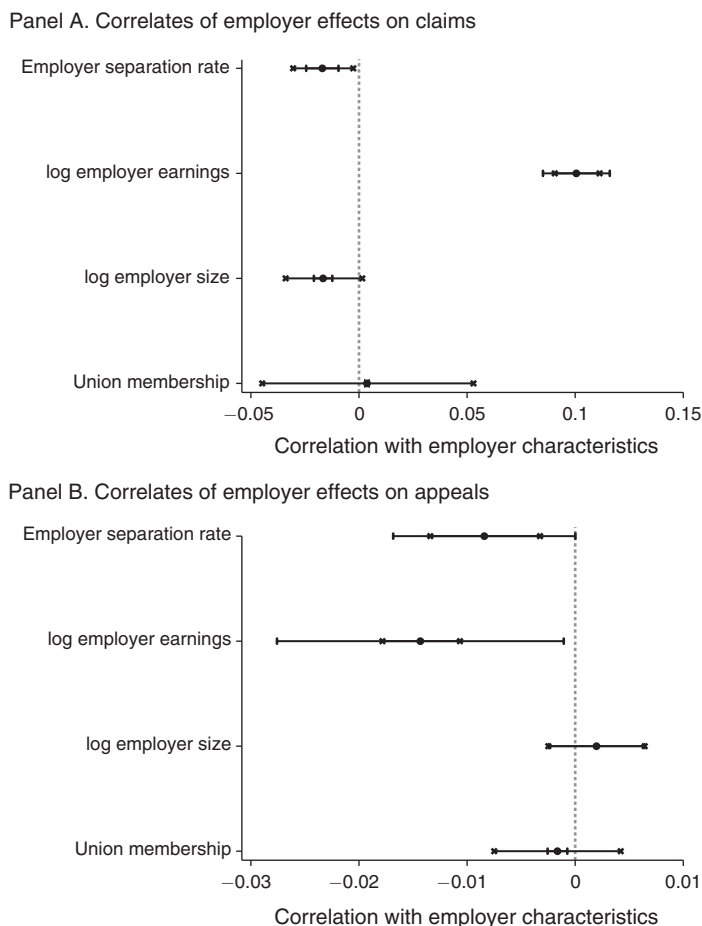


FIGURE 6. EMPLOYER-LEVEL CORRELATES OF EMPLOYER EFFECTS ON CLAIMS AND APPEALS

Notes: The top panel shows estimated coefficients (solid dots) and associated 95 percent confidence intervals (bars) from separate univariate regressions of estimated employer claim effects on various correlates. Each correlate has been transformed to a z -score, so each coefficient is interpreted as the expected change in the employer claim effect associated with a 1 standard deviation change in the correlate. The bottom panel shows estimated coefficients and confidence intervals from univariate regressions of estimated employer appeal effects on correlates. Confidence intervals denoted by |—markers are based on KSS-corrected standard errors; confidence intervals denoted by ×—markers are clustered by employer (or by sector in the case of union membership). The employer separation rate is computed using the separation rate defined in column 3 of Table 1. Unionization is calculated as a sector average for full-time workers in Washington state using the 2005–2013 CPS Monthly Outgoing Rotation Groups.

The correlations between the four employer-level characteristics and employer effects on appeals (in panel B of Figure 6) tend to mirror the correlations with claim effects. Higher employer appeal effects—a disamenity—are negatively correlated with employer earnings, positively correlated with employer size (albeit imprecisely), and negatively correlated with sector-level unionization (also imprecisely). The anomaly is that the employer appeal rate is negatively associated with employer separation rates.

In summary, employer effects on UI claims and appeals are correlated with other employer characteristics in ways that suggest high-claim and low-appeal employers are desirable.

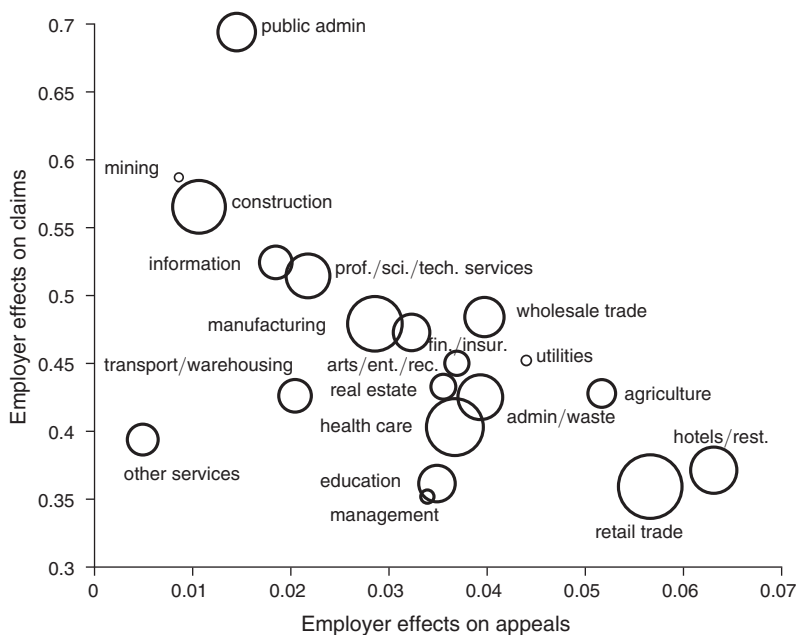


FIGURE 7. EMPLOYER EFFECTS ON CLAIMS AND APPEALS, BY SECTOR

Notes: The figure plots the sector average employer effects on claims against the sector average employer effects on appeals. The size of each circle is proportional to the number of separators in that sector, defined according to column 4 of Table 1.

B. The Plausibility of a Deterrent Effect of Appeals: Cross-Sectional Evidence

We have documented that there are employer effects on claims and appeals. In this subsection, we show that they are negatively related and suggest that part of this relationship plausibly reflects a deterrent effect of appeals on claiming.

Figure 7 plots industry averages of the employer effects on claims against employer effects on appeals. Consistent with deterrent effects, the figure shows a negative relationship. Industries with high claim and low appeal rates include mining, public administration, and construction. Industries with low claim and high appeal rates include education, retail trade, and accommodation and food services. Industries with higher claim and lower appeal rates also tend to be more unionized. The elasticity of employer claims with respect to employer appeals at the employer level is -0.53 ; in levels, each appeal is associated with about 7 fewer claims.⁴⁴

Why is it plausible that part of this negative relationship would reflect deterrent effects? Ethnographic evidence from Gould-Werth (2016) suggests that some employers tell their workers at separation that, if they claim UI, then the employer will appeal the claim. More broadly, the literature on labor market networks (e.g.,

⁴⁴To calculate this elasticity, we estimate a weighted regression of employer claim effects on employer appeal effects and convert the KSS-corrected slope from this regression to an elasticity at the sample means of worker-level claim and appeal rates. We weight observations by the number of likely eligible separating workers. While not directly comparable because the notion of an appeal is different, the only other estimate of a related elasticity comes from Anderson and Meyer (2000), which implies a somewhat lower elasticity: between -0.128 and -0.279 . See Supplemental Appendix C for details.

Caldwell and Harmon 2019 and Eliason et al. 2023) shows that workers keep in contact with past coworkers. Hence, employers' treatment of past workers could affect current workers' expectations of whether their claims might be appealed.

Employer effects on claims and appeals are bundled treatments reflecting a variety of factors (such as the employer characteristics in Figure 6). For example, some of the negative correlation could reflect other factors such as higher union coverage, which plausibly generates both higher claims and lower appeal rates. The fact that we observe an employer action (appeals) that may influence worker behavior (claiming) suggests that employer effects on claims may partly reflect worker reactions to employer behavior. In Supplemental Appendix D, we provide cross-sectional evidence linking experience rating to claims and appeals.

C. Employer Effects Partially Explain the Wage Gradient in Claims and Appeals

In Section IIF, we showed that claim and appeal rates are both highly correlated with the base-period hourly wage. In Section IVA, we showed that higher-paying employers tend to have larger employer effects on claiming. Moreover, in general, workers are sorted across employers. In this section, we quantify the extent to which sorting of workers to employers by wage explains these gradients.

How could employer effects explain (part of) the wage gradient? Loosely, the premise is that low- and high-wage workers are sufficiently sorted across employers such that differences in employer effects explain the wage gradient. But if that is the case, then how can we estimate employer effects and be confident that they apply to both high- and low-wage workers?

The following stylized example illustrates how employer effects can be identified even if workers are sorted across employers on the basis of wages. Suppose there are three employers (A, B, and C) and two types of workers (low- and high-wage). Low-wage workers move between employers A and B, and high-wage workers move between employers B and C. What pattern of claiming behavior would suggest that employer effects explain (some) of the wage gradient? If low-wage workers have higher claim rates when they separate from employer B than from employer A, while high-wage workers have higher claim rates when they separate from employer C than employer B, then we can infer that C's employer effect is higher than B's, which in turn is higher than A's. As a result, we can infer employer effects even though workers are sorted across employers by wage rate.

Panels A and B of Figure 3 showed strong wage gradients in worker-level UI claim and appeal rates. Panels A and B of Figure 8 show that the wage gradients *in employer effects* on UI claims and appeals are similar. To construct Figure 8, we assign each worker the employer effect estimated using equation (1), so we control for worker-specific factors like hourly wages. We then plot these employer effects against deciles of base-period hourly wages.⁴⁵ Employer effects on claims are strongly increasing in wages up to about \$20 an hour, a pattern also seen at the worker level. Similar to the worker-level analysis, employer effects on appeals are decreasing in wages.

⁴⁵Formally, we estimate $\psi_{ijt} = \beta_0 + \omega_{k(i)} + \epsilon_{ijt}$, where ω_k are fixed effects for the base-period hourly wage decile of worker i , and we omit the first decile.

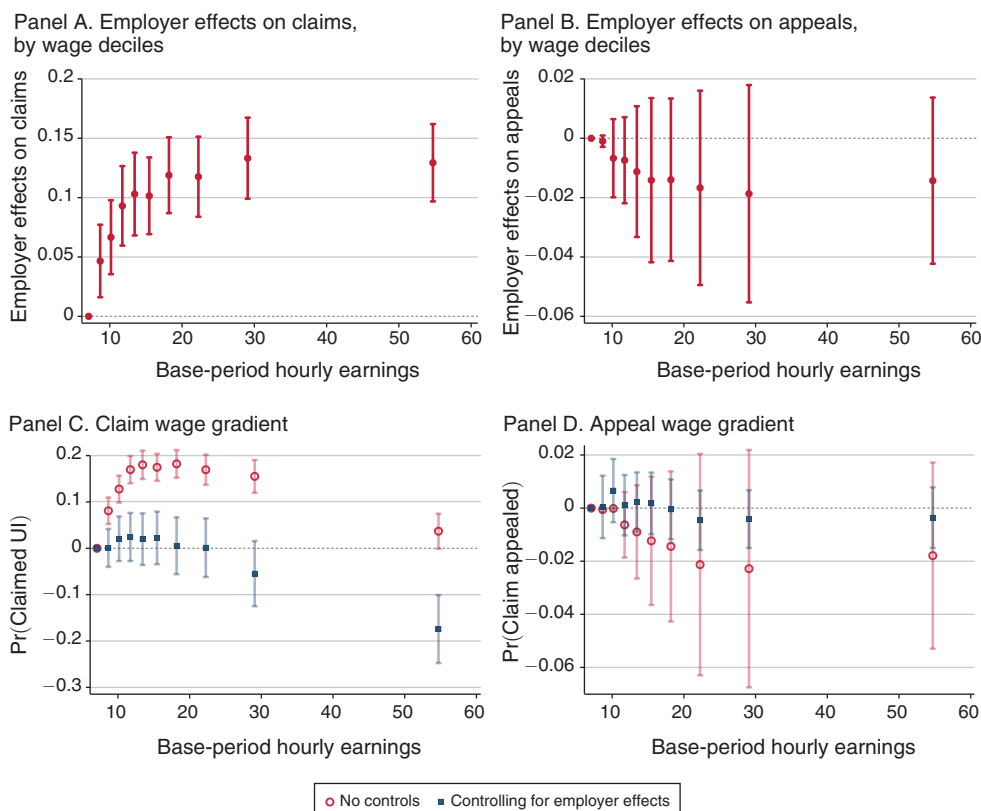


FIGURE 8. EMPLOYER EFFECTS ON CLAIMS AND APPEALS, BY WORKER-LEVEL HOURLY EARNINGS

Notes: Panel A plots coefficients from column 3 of Supplemental Appendix Table A5, showing the relationship between estimated employer effects on claims and deciles of worker base-period hourly earnings. Panel B plots coefficients from column 7 of Supplemental Appendix Table A5, showing the relationship between estimated employer effects on appeals and deciles of worker hourly earnings. Panel C plots coefficients from columns 1 and 2 of Supplemental Appendix Table A5. The hollow red circles plot the relationship between worker claim rates and deciles of hourly earnings. The blue squares plot the same relationship, after controlling for estimated employer effects on claims. Panel D repeats this analysis for appeal rates by plotting the coefficients from columns 5 and 6 of Supplemental Appendix Table A5. The hollow red circles plot the relationship between appeal rates and deciles of hourly earnings. The blue squares represent the relationship between appeal rates and hourly earnings deciles, after controlling for estimated employer effects on appeals.

Panel C in Figure 8 shows the significant role of employer effects in explaining the wage gradient in claiming. The figure plots two data series. The first is the mean claim rate by wage decile. The second series is the same decile means, after controlling for employer effects on claiming.⁴⁶ The wage gradient from the first to the fifth decile is reduced by almost 90 percent when we control for employer effects.⁴⁷

⁴⁶ Because employer effects are noisily estimated, we pursue a split-sample approach analogous to what we did in Table 3 (column 2) and instrument one estimate of the employer effect with the other.

⁴⁷ In Supplemental Appendix Table A5, we present the regression versions of these figures. Column 1 shows the decile means of claim rates, column 2 shows the decile means of claim rates when we control for the employer effects (and instrument the employer effects using a split-sample approach), and column 3 shows the employer effects by decile. Comparing columns 1 and 2, we can see that the fact that the coefficient on the employer effects is approximately 1 means that the difference is close to the employer effects by decile in column 3.

Panel D in Figure 8 shows the significant role of employer effects in explaining the wage gradients in appeals. The construction of panel D parallels that of panel C, except that in panel D we assume the coefficient on the employer effects on appeals is 1, rather than estimating it. The reason is that employer effects on appeals are noisily estimated: The split-sample IV approach gives a first-stage F -statistic close to zero, so the point estimates are not reliable (see Supplemental Appendix Table A2). The wage gradient from the first to the fifth decile is largely eliminated when we adjust for employer effects.⁴⁸ In summary, employer effects play a large role in explaining wage gradients in both UI claims and appeals.

D. Progressivity of the UI System

Given that employer effects explain a significant portion of the wage gradient in claiming, we might wonder about their distributional consequences. In this section, we show that equalizing employer effects across workers would increase the progressivity of the system.

We first compute the progressivity of the existing system. This calculation compares the benefits received by each worker with the taxes paid per year on their behalf, which is the taxable portion of the worker's earnings multiplied by their employer's UI tax rate.⁴⁹ At the worker level, this calculation is noisy because receipt of UI in a given year is a low-probability event.⁵⁰ To address this issue, we aggregate individuals into centiles based on their average hourly wage rate for the entire time they are in the sample, which comes closer to approximating their permanent income. We pool the years in the sample (2005 to 2013), reporting annual averages of taxes paid and benefits received of the workers in each centile, weighted by the number of years each worker appears in the sample. Supplemental Appendix E provides details.

Panel A of Figure 9 illustrates the progressivity of the existing system using Lorenz curves of both UI taxes and benefits (the cumulative share of UI taxes paid and benefits received sorted by centiles of the hourly wage distribution).⁵¹ The curves for both taxes (dotted red line) and benefits (solid black line) lie below the 45-degree line, so higher-wage workers pay more in UI taxes and receive more in UI benefits. The Lorenz curves for taxes and benefits lie approximately on top of each other up to the median of the wage distribution, after which the benefit curve rises faster than the tax curve. Hence, the redistribution in Washington's UI system from 2005 to 2013 occurred mainly within the upper half of the wage distribution. Table 5 displays various statistics related to the Lorenz curves. Panel A shows that the Gini coefficient is indeed higher for taxes than for benefits, so the system is on net progressive. Consistent with the figure, panel B shows that the shares of taxes paid

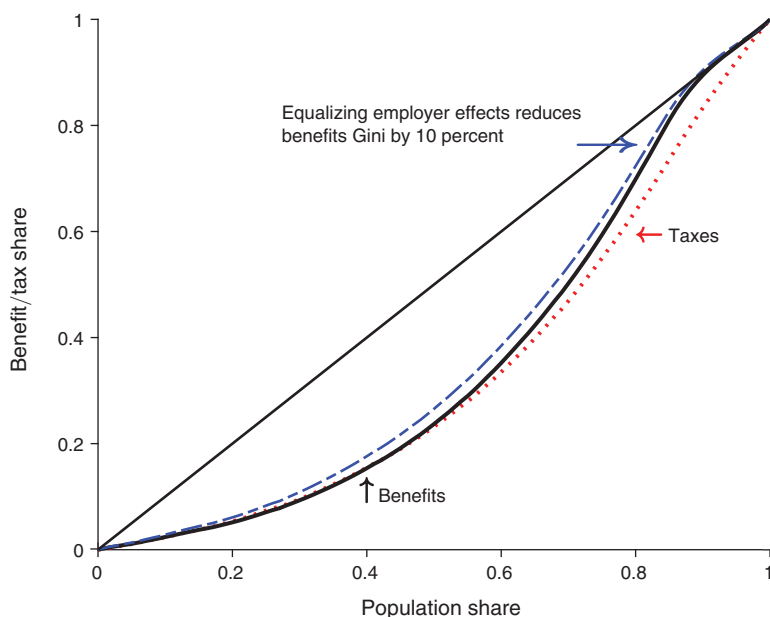
⁴⁸ See Supplemental Appendix Table A5 for the numbers underlying this figure.

⁴⁹ This calculation assumes complete pass-through of UI taxes to worker wages at the individual worker level; this assumption is consistent with the evidence for market-wide changes in UI taxes documented in Anderson and Meyer (1997a).

⁵⁰ For example, in Table 1, the probability of UI receipt among all workers is about 1.5 percent.

⁵¹ Supplemental Appendix Figure A7, panel A shows the levels of taxes and benefits underlying Figure 9. The closest precedent for our analysis is Guo and Johnston (2021, Figure 6) for Florida in 2005 to 2006. Supplemental Appendix Figure A7, panel B shows this figure for each year. The patterns are very similar to those reported by Guo and Johnston (2021).

Panel A. Lorenz curves of UI taxes and benefits



Panel B. UI claim rates, by hourly wages

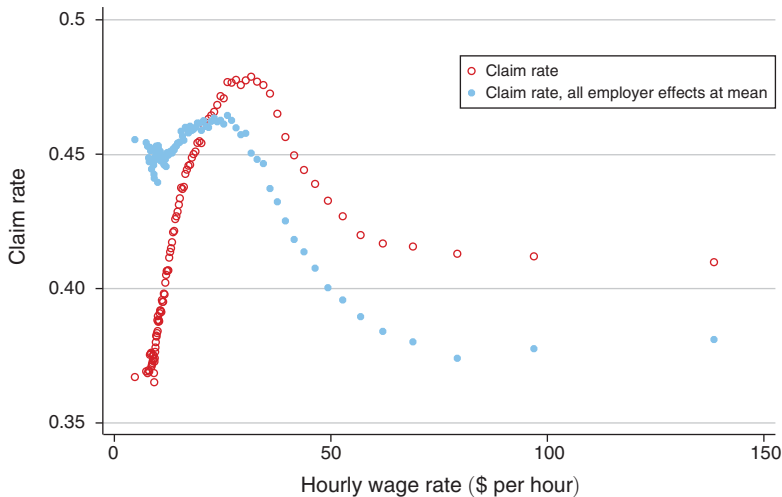


FIGURE 9. PROGRESSIVITY OF UI TAXES AND BENEFITS

Notes: Panel A plots the cumulative share of UI taxes paid and benefits received by share of the sample, sorted by centiles of hourly earnings; see Table 5. The dotted red line plots UI taxes paid on behalf of workers, the solid black line plots UI benefits received, and the dashed blue line plots the UI benefits received in a counterfactual where employer effects on claiming are equalized across all employers to the sample mean. In Panel B, the hollow red dots show predicted claim rates for each centile, and the solid blue dots show simulated claim rates for each centile, where the employer effects on claiming have been set to the sample mean; see Supplemental Appendix E for details.

and benefits received are surprisingly similar for above- and below-median-wage workers. Panel C shows that the redistribution occurs mainly from the top decile to the fortieth to eightieth percentiles of the wage distribution, with the biggest net beneficiary being the eighth decile (seventieth to eightieth percentiles).

TABLE 5—PROGRESSIVITY OF UI TAXES AND BENEFITS

	Taxes (1)	Benefits (2)	Benefits: equalizing employer effects (3)
<i>Panel A. Gini coefficients</i>			
	0.335	0.300	0.264
<i>Panel B. Share by below-/above-median hourly wage</i>			
Below	0.232	0.237	0.266
Above	0.768	0.763	0.734
<i>Panel C. Share by hourly wage decile</i>			
1	0.024	0.024	0.028
2	0.030	0.028	0.033
3	0.042	0.042	0.048
4	0.060	0.061	0.068
5	0.077	0.083	0.089
6	0.103	0.115	0.119
7	0.134	0.150	0.150
8	0.170	0.196	0.189
9	0.194	0.196	0.181
10	0.167	0.105	0.096

Notes: This table analyzes the progressivity of the UI system. Column 1 displays distributional statistics of the UI payroll tax paid by each worker under the assumption that the incidence of the tax falls fully on the worker. Column 2 displays distributional statistics of the UI benefits received by each worker. Column 3 shows these distributional statistics in a counterfactual where the employer effects on claiming are equalized across all employers. In all cases, each worker is assigned to an hourly wage centile based on the worker's average wage over all years the worker is observed (see Supplemental Appendix E for details). Panel A displays Gini coefficients of taxes paid, benefits received, and benefits after equalizing employer effects. Panel B displays the share of taxes paid and benefits received among above- and below-median hourly wage workers. Panel C displays the share of taxes paid and benefits received by hourly wage decile.

How would the progressivity of UI change if all employers had the average employer effect on claims (that is, if we changed only the claim rate, not the tax rate)? Panel A of Figure 8 showed a steep wage gradient in employer effects on claiming, and the analysis in the previous section showed that employer effects accounted for about half of that wage gradient. Hence, we expect equalizing employer effects on claiming across all workers to increase the progressivity of the system.

Panel B of Figure 9 shows the change in worker claiming when employer effects are equalized. (Supplemental Appendix E provides details.) The resulting benefit curve (dashed blue line) in panel A now diverges from the tax curve starting at the bottom of the wage distribution, such that the UI system is more redistributive, shifting benefits toward the bottom half of the wage distribution. Panel A of Table 5 shows that the Gini coefficient for benefits declines by more than 10 percent, and panel B shows that the share of benefits received by the bottom half of the wage distribution increases by about 3 percentage points (more than 10 percent). Panel C shows that this redistribution is achieved by making workers in the ninth decile of the wage distribution net payers into the system.

V. Claim, Appeal, and Receipt Rates among Eligible and Ineligible Workers

In this section, we measure claim rates, appeal rates, and receipt rates (conditional on appeals) among eligible and ineligible separators. We use the employer contraction as an instrument that shifts the composition of separating workers toward UI eligibility. We use these estimates to quantify the targeting properties of UI and show that it is plausible that small reductions in experience rating would reduce targeting errors.⁵²

A. Methods

Our approach uses an employer's contraction rate as an instrument for the eligibility composition of separators, where the idea is that when an employer contracts, the marginal separators are likely to be separation eligible for UI. Suppose we measure outcomes at two growth rates: a no-growth region (a neighborhood of growth rates around a 0 percent growth rate) and a mass-layoff region (a neighborhood around a 15 percent contraction). Denote these two growth rates by $g \in \{ng, ml\}$, and suppose we observe the following outcomes at each growth rate: a separation rate (sep_g), a claim rate (cl_g), a probability that a claim is appealed (pa_g), and a probability of UI receipt conditional on appeal (rec_g).

Now assume that the observed outcomes represent a mixture of the outcomes among two types of workers: those who are ineligible and eligible, denoted by $e \in \{0, 1\}$. (We restrict attention to monetarily eligible separators, so the relevant notion of eligibility is whether a worker is separation eligible.) Denote by C_e , p_e , and r_e the claim rate, the appeal rate, and the receipt rate conditional on appeal, for workers of eligibility type e .

Consistent with the evidence in Section IIC (and Figure 2), we assume that a share σ of workers who separate in the no-growth region are eligible for UI. We then assume that, when an employer contracts, the incremental separators have lost jobs due to lack of work and meet the separation eligibility requirements.

In Supplemental Appendix F we show that, given rates of separations, claims, appeals, and receipt at the two growth rates, we can identify the claim, appeal, and receipt rates among the eligible separators. Heuristically, employer contraction is an instrument for eligible separations. Since the eligible separators can be interpreted as the compliers, Wald-like estimators recover these rates among the eligible. To do so among the ineligible, we need one additional moment: the overall share of UI recipients who are ineligible. In Supplemental Appendix F, we discuss why we need this moment and the estimate we use.

To estimate the separation, claim, appeal, and conditional-on-appeal receipt rates, we analyze outcomes as a function of employer growth rates, using 21 bins of employer growth rates. Each bin is 5 percentage points wide, with one bin centered on 0 and the remaining bins ranging from 52.5 percent contraction

⁵²We restrict attention to the sample of likely eligible separators in column 4 of Table 1. If we instead used the sample of all transitions in column 3 of Table 1, which includes, for example, workers who made job-to-job transitions or left the labor force, then our instrument would fail. As Flaaen, Shapiro, and Sorkin (2019) show, employer contraction induces voluntary as well as involuntary separations.

to 52.5 percent expansion. We present specifications with and without employer fixed effects; however, our preferred specification uses within-employer variation in growth rates. In Supplemental Appendix Figure A8, we present the resulting estimates, and in Supplemental Appendix F.3, we discuss their interpretation. To recover the parameters of the model (C_e , p_e , and r_e), we need to pick a growth rate to use as the contraction region. We select the employer growth bin centered on a 15 percent contraction because it represents a substantial contraction without greatly changing an employer's marginal incentives to appeal claims. (In Supplemental Appendix F.3 we discuss how an employer's incentives to appeal vary with the size of a contraction, show how the statistical model fits the data at other growth rates, and show that most of our estimates are not particularly sensitive to the choice of growth rate used.)

B. Targeting Properties of UI

The targeting properties of social programs are central to program analysis: Are those who receive the program's benefits the intended recipients? Are those who do not have access not supposed to have access? Although these questions have been addressed for other programs, to our knowledge, the targeting properties of UI have not previously been quantified.⁵³

The key inputs into the targeting estimates are the behavior of eligible and ineligible separators. The first two columns of Table 6 report the baseline parameter estimates.⁵⁴ The claim rate among eligible separators is 57.5 percent, which is about 4 times the rate among ineligible separators (16.5 percent). The appeal rate of ineligible claimants is over 4 times that of eligible claimants (10.2 percent versus 2.2 percent), and the receipt rate conditional on appeal of eligibles is about twice that of ineligibles (74.3 percent versus 40.2 percent). Finally, nearly two-thirds of separators among employers in the no-growth region are separation eligible for UI.⁵⁵

Following Kleven and Kopczuk (2011), we combine these estimates to distinguish among three types of targeting errors and report each in Table 6. First, Type IA errors are false negatives stemming from eligible separators not claiming: About 28 percent of separators are in this category. Second, Type IB errors are false negatives stemming from eligible separators claiming but not receiving benefits (because their claim was appealed): Only 0.2 percent of separators are in this category. Finally, Type II errors are false positives stemming from ineligible separators claiming and receiving benefits: About 5 percent of separators are in this category.

⁵³ Some areas and papers are housing assistance (Reeder 1985), Supplemental Security Income and disability insurance (Benitez-Silva, Buchinsky, and Rust 2004), disability insurance (Low and Pistaferri 2015 and Deshpande and Li 2019), food stamps (Finkelstein and Notowidigdo 2019), and Medicaid home health care (Lieber and Lockwood 2019). Some of these studies look at targeting in terms of the marginal utility of consumption; we instead look at targeting relative to program rules, which is also of interest to policymakers.

⁵⁴ The moments we use come from the bins centered on a 0 percent growth rate ("no growth") and a 15 percent contraction ("mass layoff") with employer fixed effects in Supplemental Appendix Table A6. Because the constant in the fixed effect regressions is not identified, we renormalize the values so that in the fixed effect regression the values match OLS estimates at zero.

⁵⁵ In Supplemental Appendix G, we conduct a validation exercise by reestimating this model on subsets of the data to show that claim and appeal rates among eligible separators align with the estimated AKM employer effects on claims and appeals. We also show that claim and appeal rates among eligible separators follow a wage gradient similar to the overall claim and appeal rates.

TABLE 6—MODEL ESTIMATES OF UI OUTCOMES FOR ELIGIBLE AND INELIGIBLE SEPARATORS

	Baseline estimates		Counterfactual: Experience rating reduced by 10%	
	Eligibles	Ineligibles	Eligibles	Ineligibles
Elasticity of claiming to appeals			−0.53	
Elasticity of appeals to experience rating			1.607	
	(1)	(2)	(3)	(4)
Panel A. Baseline estimates				
Estimated rates				
Claim rate (C_e)	0.575	0.165	0.624	0.179
Appeal rate claim (p_e)	0.022	0.102	0.019	0.086
Receipt rate appeal (r_e)	0.743	0.402	0.743	0.402
Targeting errors				
Type IA (eligible, no claim)	0.281		0.249	
Type IB (eligible, claim, no receipt)	0.002		0.002	
Type II (ineligible, receive)	0.052		0.057	
Total targeting errors	0.336		0.308	
Counterfactual: Alternative elasticities				
Elasticity of claiming to appeals	−0.128		−0.128	
Elasticity of appeals to experience rating	0.804		3.214	
	Eligibles	Ineligibles	Eligibles	Ineligibles
	(5)	(6)	(7)	(8)
			Eligibles	Ineligibles
			(9)	(10)
			Eligibles	Ineligibles
			(11)	(12)
Panel B. Alternative choices of elasticities				
Estimated rates				
Claim rate (C_e)	0.581	0.167	0.599	0.172
Appeal rate claim (p_e)	0.021	0.094	0.015	0.070
Receipt rate appeal (r_e)	0.743	0.402	0.743	0.402
Targeting errors				
Type IA (eligible, no claim)	0.277		0.266	0.273
Type IB (eligible, claim, no receipt)	0.002		0.002	0.002
Type II (ineligible, receive)	0.053		0.056	0.054
Total targeting errors	0.333		0.323	0.329

Notes: This table displays estimates from the statistical model in Section V. At employers that are neither growing nor shrinking, 66.2 percent of likely eligible separators are estimated to be actually eligible ($e = 1$), and 33.8 percent are estimated to be ineligible ($e = 0$). Columns 1 and 2 show various rates for each of these groups. Columns 3 and 4 show the results of an analysis where experience rating is reduced by 10 percent, using our baseline elasticities of claiming with respect to appeals and appeals with respect to experience rating. Panel B considers the sensitivity of this analysis to alternative choices of the elasticities.

Two important conclusions follow. First, targeting errors are widespread: About one-third of separators incorrectly receive or do not receive UI. Second, the main source of targeting errors in UI stems from eligible separators who do not claim.

C. Counterfactual Analysis of a Change in Experience Rating

We now consider how a change in experience rating might affect the claim rate and targeting properties of UI. To do the analysis, we need two elasticities: the elasticity of claims with respect to appeals and the elasticity of appeals with respect to experience rating.⁵⁶

⁵⁶This analysis ignores any effect of changing experience rating on the behavior of workers outside the sample of likely eligible workers. In addition, this analysis represents a partial assessment of reducing experience rating in that it does not take into account any changes in layoff behavior of employers.

We take the first elasticity from Section IVB, where the point estimate in our data using cross-sectional variation is -0.53 . This estimate overstates the true elasticity if it reflects noncausal factors such as higher union coverage leading to lower appeal rates and higher claim rates. Accordingly, we examine the sensitivity of the results using estimates from Anderson and Meyer (2000), which range from -0.13 to -0.28 .⁵⁷

We derive the second elasticity by comparing the estimated appeal rates for eligible and ineligible separators with the estimated receipt rates for eligible and ineligible separators; see Supplemental Appendix F.4 for the derivation. The idea is that there is variation in the data in the extent to which workers are experience rated. An ineligible claimant is much less likely than an eligible claimant to receive UI, so in expectation employers save more in UI taxes when they appeal the claims of ineligible than eligible workers. To the extent employers can correctly infer the eligibility status of their separating workers, this comparison identifies the elasticity of the appeal rate with respect to marginal changes in experience rating. This comparison gives an elasticity of 1.6. We also check the sensitivity of the results to doubling and halving this estimate.

Columns 3 and 4 of Table 6 show how a 10 percent decrease in experience rating might affect take-up and targeting under the assumptions that the elasticity of claims with respect to appeals is 1.6 and the elasticity of appeals with respect to experience rating is -0.53 . (See Supplemental Appendix F.6 for details.) In all cases, the claim rates for eligible and ineligible workers both rise. This result highlights the trade-off presented by experience rating: Decreasing experience rating encourages ineligible workers to claim, which is undesirable, and encourages eligible workers to claim, which is desirable. The mechanism for this change is that reduced experience rating reduces the incentive for employers to appeal claims.⁵⁸

Table 6 also suggests that decreasing experience rating would increase targeting efficiency. Mechanically, the model finds that more eligible than ineligible workers claim, so increasing the claim rate of both groups by the same proportion results in a larger increase in the number of claims by eligible workers, which in turn decreases targeting errors. This qualitative result is quite robust. Undoing it would require the responsiveness of claims to experience rating to be about seven times higher for ineligible workers than eligible workers.⁵⁹ Thus, if policymakers value different types of targeting errors equally, then decreasing experience rating would be desirable.

The remaining columns of Table 6 present a sensitivity analysis where we vary the relevant elasticities over the upper and lower bounds described above. Although the magnitudes move somewhat (and in predictable ways), in all cases, reducing experience rating reduces targeting errors.

⁵⁷ Appeals and separation-issue denials are similar: Anderson and Meyer's (2000) elasticities range from -0.128 to -0.279 . See Supplemental Appendix C for details of this calculation.

⁵⁸ Conditional on a claim being appealed, the probability that a worker receives benefits remains unchanged.

⁵⁹ Table 6 implies that there are about seven times as many claims from eligible than ineligible workers: $(0.575 \times 0.662) / (0.165 \times 0.338) \approx 7$.

VI. Summary and Conclusion

The UI payroll tax is experience rated at the level of the employer, and each employer's UI payroll tax rate is based on the "charging" of benefits paid to workers laid off by that employer. UI take-up is far from complete, and there are steep income gradients in both worker claiming and employer appeals. Specifically, low-wage workers are less likely to claim and more likely to have their claims appealed than median-wage workers. Thus, UI take-up is both incomplete and unequal over the wage distribution.

We examine whether these relationships depend on factors relating to the worker or the employer by decomposing the variation in UI claims and appeals into worker and employer effects. We find that employer effects on both claiming and appealing are highly dispersed; for both, 1 standard deviation of the employer effect is at least one-third of the sample mean, and the relative variance of employer effects to worker effects is larger for both of these outcomes than for wages.

Consistent with employer appeals deterring workers from claiming, employer effects on appeals and claims are negatively correlated. We show that high-claim and low-appeal employers are higher paying. Moreover, we find that well over half of the income gradients can be explained by employer effects. Thus, equalizing employer effects across the wage distribution would increase the progressivity of the UI system.

A statistical model of UI take-up shows that the dominant source of UI targeting error is that not all eligible workers claim benefits. We link this incomplete claiming to the deterrent effects of employer appeals, which ultimately comes from the experience rating of the UI payroll tax. Decreases in experience rating would likely reduce the extent of targeting errors. In summary, the paper highlights a novel dimension of the role of employers in the labor market and emphasizes a little-studied factor affecting the take-up of UI.

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